

Statistical_FDS_Assignment_1_Solutions

November 22, 2025

1 Statistical Foundation of Data Science - Assignment 1 Solutions

Course Name: Statistical Foundation of Data Science

Course Code: CSU 1658

Max Points: 50

This notebook contains step-by-step solutions to all assignment problems.

1.1 Environment Setup and Dependencies

Before executing this notebook, ensure all required packages are installed in your virtual environment. This section will install and verify all necessary dependencies for statistical analysis and data visualization.

1.1.1 Required Dependencies:

- **pandas:** Data manipulation and analysis library
- **numpy:** Numerical computing library for arrays and mathematical functions
- **matplotlib:** Comprehensive plotting library for creating static visualizations
- **seaborn:** Statistical data visualization library built on matplotlib
- **scipy:** Scientific computing library containing statistical functions and tests
- **jupyter:** For notebook environment (if not already installed)

Run the following cell to install all dependencies in your virtual environment:

```
[16]: # Dependency Installation and Verification
import subprocess
import sys
import pkg_resources

def install_package(package):
    """Install a package using pip if not already installed"""
    try:
        pkg_resources.get_distribution(package)
        print(f"[CHECK] {package} is already installed")
    except pkg_resources.DistributionNotFound:
        print(f"Installing {package}...")
        subprocess.check_call([sys.executable, "-m", "pip", "install", package])
        print(f"[CHECK] {package} installed successfully")
```

```

# List of required packages
required_packages = [
    'pandas>=1.3.0',
    'numpy>=1.21.0',
    'matplotlib>=3.4.0',
    'seaborn>=0.11.0',
    'scipy>=1.7.0',
    'jupyter>=1.0.0'
]

print("Installing and verifying dependencies...")
print("=" * 50)

for package in required_packages:
    package_name = package.split('>=')[0] # Extract package name without version
    install_package(package_name)

print("\n" + "=" * 50)
print("All dependencies installed successfully!")
print("You can now proceed with the assignment solutions.")

# Verify installations by importing
try:
    import pandas as pd
    import numpy as np
    import matplotlib.pyplot as plt
    import seaborn as sns
    from scipy import stats
    print("\n[CHECK] All imports successful - Ready to proceed!")
except ImportError as e:
    print(f"\n[CROSS] Import error: {e}")
    print("Please restart the kernel and try again.")

```

```

Installing and verifying dependencies...
=====

pandas is already installed
numpy is already installed
matplotlib is already installed
seaborn is already installed
scipy is already installed
jupyter is already installed

=====
All dependencies installed successfully!
You can now proceed with the assignment solutions.

All imports successful - Ready to proceed!

```

```
[17]: # Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from scipy.stats import chi2_contingency, pearsonr, spearmanr
import warnings
warnings.filterwarnings('ignore')

# Set display options for better output
pd.set_option('display.max_columns', None)
pd.set_option('display.width', None)
pd.set_option('display.max_colwidth', None)
```

1.2 Very Short Questions ($5 \times 2 = 10$ marks)

1.2.1 Q1. Explain the different types of Data Structures in Pandas.

Answer:

Pandas provides two fundamental data structures that form the backbone of data analysis in Python. Understanding these structures is crucial for effective data manipulation and analysis.

1.3 Primary Data Structures in Pandas

1.3.1 1. Series (One-Dimensional Data Structure)

A **Series** is a one-dimensional labeled array capable of holding any data type including integers, strings, floating-point numbers, Python objects, and more.

Key Characteristics: - **Homogeneous Data:** All elements in a Series must be of the same data type - **Labeled Index:** Each element has an associated label (index) for easy access - **Size Immutable:** Once created, the size cannot be changed (though values can be modified) - **Vector Operations:** Supports vectorized operations and broadcasting

Mathematical Representation: If we have a Series S with n elements, it can be represented as:

$$S = \{(i_1, v_1), (i_2, v_2), \dots, (i_n, v_n)\}$$

Where: - $i_1, i_2, \dots, i_n$ are the index labels - $v_1, v_2, \dots, v_n$ are the corresponding values

Use Cases: - Time series data (stock prices over time) - Single column of data from a dataset - Results of statistical calculations - Intermediate results in data processing pipelines

1.3.2 2. DataFrame (Two-Dimensional Data Structure)

A **DataFrame** is a two-dimensional labeled data structure with columns of potentially different types. It is the most commonly used pandas object and resembles a spreadsheet or SQL table.

Key Characteristics: - **Heterogeneous Data:** Different columns can contain different data types - **Row and Column Labels:** Both axes (rows and columns) have labeled indices - **Size**

Mutable: Columns can be added, deleted, or modified - **Tabular Structure:** Represents data in a table format with rows and columns

Mathematical Representation: A DataFrame D with m rows and n columns can be represented as:

$$D = \{(r_i, c_j, v_i_j) \mid i \in [1, m], j \in [1, n]\}$$

Where: - r_i represents the row index - c_j represents the column index
- v_i_j represents the value at position (i,j)

Use Cases: - Complete datasets with multiple variables - Database query results - Spreadsheet-like data manipulation - Statistical analysis requiring multiple columns

1.3.3 Comparison Summary

	Aspect	Series	DataFrame
Dimensions	1D		2D
Data Types	Homogeneous		Heterogeneous (per column)
Index	Single index		Row and column indices
Structure	Vector-like		Table-like
Memory Usage	Lower		Higher
Complexity	Simple		Complex

1.3.4 Practical Demonstration

The following code examples illustrate the creation and basic operations of both data structures:

```
[18]: # Comprehensive Demonstration of Pandas Data Structures

print("PANDAS DATA STRUCTURES DEMONSTRATION")
print("=" * 60)

# =====
# PART 1: PANDAS SERIES DEMONSTRATION
# =====
print("\n1. PANDAS SERIES EXAMPLES")
print("-" * 40)

# Example 1: Creating a Series with explicit index
print("Example 1: Series with Custom Index")
temperature_data = [22.5, 25.1, 23.8, 26.2, 24.9]
cities = ['New York', 'London', 'Tokyo', 'Sydney', 'Mumbai']
temperature_series = pd.Series(temperature_data, index=cities,
                               name='Temperature_Celsius')

print("Temperature Series:")
print(temperature_series)
print(f"\nSeries Type: {type(temperature_series)}")
```

```

print(f"Data Type: {temperature_series.dtype}")
print(f"Series Name: {temperature_series.name}")
print(f"Index: {list(temperature_series.index)}")
print(f"Values: {list(temperature_series.values)}")

# Example 2: Series with automatic index
print("\n" + "-" * 30)
print("Example 2: Series with Default Index")
sales_data = [1500, 2300, 1800, 2100, 1950]
sales_series = pd.Series(sales_data, name='Monthly_Sales')
print("Sales Series:")
print(sales_series)

# Example 3: Series operations
print("\n" + "-" * 30)
print("Example 3: Series Operations")
print(f"Mean temperature: {temperature_series.mean():.2f}°C")
print(f"Maximum temperature: {temperature_series.max():.2f}°C")
print(f"City with highest temperature: {temperature_series.idxmax()}")

# Statistical summary
print("\nStatistical Summary:")
print(temperature_series.describe())

print("\n" + "=" * 60)

# =====
# PART 2: PANDAS DATAFRAME DEMONSTRATION
# =====
print("\n2. PANDAS DATAFRAME EXAMPLES")
print("-" * 40)

# Example 1: Creating DataFrame from dictionary
print("Example 1: DataFrame from Dictionary")
employee_data = {
    'Employee_ID': [101, 102, 103, 104, 105],
    'Name': ['Alice Johnson', 'Bob Smith', 'Charlie Brown', 'Diana Ross', 'Edward Wilson'],
    'Department': ['Engineering', 'Marketing', 'Engineering', 'Sales', 'HR'],
    'Salary': [75000, 62000, 78000, 58000, 65000],
    'Years_Experience': [5, 3, 7, 2, 4],
    'Performance_Score': [4.2, 3.8, 4.5, 3.9, 4.1]
}

employee_df = pd.DataFrame(employee_data)
print("Employee DataFrame:")
print(employee_df)

```

```

print(f"\nDataFrame Type: {type(employee_df)}")
print(f"Shape (rows, columns): {employee_df.shape}")
print(f"Column Names: {list(employee_df.columns)}")
print(f"Index: {list(employee_df.index)}")

# Example 2: DataFrame information and statistics
print("\n" + "-" * 30)
print("Example 2: DataFrame Information")
print("\nData Types:")
print(employee_df.dtypes)

print("\nDataFrame Info:")
print(employee_df.info())

print("\nNumerical Statistics:")
print(employee_df.describe())

# Example 3: DataFrame operations
print("\n" + "-" * 30)
print("Example 3: DataFrame Operations")

# Selecting specific columns
print("Salary Statistics:")
salary_stats = employee_df['Salary'].describe()
print(salary_stats)

# Filtering data
high_performers = employee_df[employee_df['Performance_Score'] > 4.0]
print(f"\nHigh Performers (Score > 4.0):")
print(high_performers[['Name', 'Department', 'Performance_Score']])

# Grouping by department
print(f"\nAverage Salary by Department:")
dept_salary = employee_df.groupby('Department')['Salary'].mean()
print(dept_salary)

print("\n" + "=" * 60)

# =====
# PART 3: COMPARISON AND RELATIONSHIPS
# =====
print("\n3. SERIES vs DATAFRAME RELATIONSHIPS")
print("-" * 40)

# Extract a Series from DataFrame
print("Extracting Series from DataFrame:")

```

```

name_series = employee_df['Name']
print(f"Name column as Series:")
print(name_series)
print(f"Type: {type(name_series)}")

# Create DataFrame from multiple Series
print("\n" + "-" * 30)
print("Creating DataFrame from Multiple Series:")

# Create individual Series
ids = pd.Series([201, 202, 203], name='Product_ID')
products = pd.Series(['Laptop', 'Mouse', 'Keyboard'], name='Product_Name')
prices = pd.Series([999.99, 25.50, 75.00], name='Price')

# Combine into DataFrame
product_df = pd.DataFrame({'ID': ids, 'Product': products, 'Price': prices})
print("Product DataFrame from Series:")
print(product_df)

print("\n" + "=" * 60)
print("SUMMARY OF KEY DIFFERENCES:")
print("-" * 40)
print("1. Dimensionality: Series (1D) vs DataFrame (2D)")
print("2. Data Types: Series (homogeneous) vs DataFrame (heterogeneous per_
    ↳column)")
print("3. Indexing: Series (single index) vs DataFrame (row + column indices)")
print("4. Use Cases: Series (single variable) vs DataFrame (multiple variables)")
print("5. Memory: Series (lower overhead) vs DataFrame (higher overhead)")

# Memory usage comparison
print(f"\nMemory Usage Comparison:")
print(f"Temperature Series: {temperature_series.memory_usage(deep=True)} bytes")
print(f"Employee DataFrame: {employee_df.memory_usage(deep=True).sum()} bytes")

```

PANDAS DATA STRUCTURES DEMONSTRATION

=====

1. PANDAS SERIES EXAMPLES

Example 1: Series with Custom Index

Temperature Series:

New York	22.5
London	25.1
Tokyo	23.8
Sydney	26.2
Mumbai	24.9

Name: Temperature_Celsius, dtype: float64

```

Series Type: <class 'pandas.core.series.Series'>
Data Type: float64
Series Name: Temperature_Celsius
Index: ['New York', 'London', 'Tokyo', 'Sydney', 'Mumbai']
Values: [np.float64(22.5), np.float64(25.1), np.float64(23.8), np.float64(26.2),
np.float64(24.9)]

```

Example 2: Series with Default Index

Sales Series:

```

0    1500
1    2300
2    1800
3    2100
4    1950

```

Name: Monthly_Sales, dtype: int64

Example 3: Series Operations

Mean temperature: 24.50°C

Maximum temperature: 26.20°C

City with highest temperature: Sydney

Statistical Summary:

```

count    5.000000
mean     24.500000
std       1.405347
min      22.500000
25%      23.800000
50%      24.900000
75%      25.100000
max      26.200000

```

Name: Temperature_Celsius, dtype: float64

=====

2. PANDAS DATAFRAME EXAMPLES

Example 1: DataFrame from Dictionary

Employee DataFrame:

	Employee_ID	Name	Department	Salary	Years_Experience	\
0	101	Alice Johnson	Engineering	75000		5
1	102	Bob Smith	Marketing	62000		3
2	103	Charlie Brown	Engineering	78000		7
3	104	Diana Ross	Sales	58000		2
4	105	Edward Wilson	HR	65000		4

Performance_Score

0	4.2
1	3.8
2	4.5
3	3.9
4	4.1

```
DataFrame Type: <class 'pandas.core.frame.DataFrame'>
Shape (rows, columns): (5, 6)
Column Names: ['Employee_ID', 'Name', 'Department', 'Salary',
'Years_Experience', 'Performance_Score']
Index: [0, 1, 2, 3, 4]
```

----- Example 2: DataFrame Information

```
Data Types:
Employee_ID      int64
Name             object
Department       object
Salary           int64
Years_Experience int64
Performance_Score float64
dtype: object
```

```
DataFrame Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Employee_ID           5 non-null     int64
1   Name                  5 non-null     object
2   Department            5 non-null     object
3   Salary                5 non-null     int64
4   Years_Experience      5 non-null     int64
5   Performance_Score     5 non-null     float64
dtypes: float64(1), int64(3), object(2)
memory usage: 372.0+ bytes
None
```

Numerical Statistics:

	Employee_ID	Salary	Years_Experience	Performance_Score
count	5.000000	5.000000	5.000000	5.000000
mean	103.000000	67600.000000	4.200000	4.100000
std	1.581139	8561.541917	1.923538	0.273861
min	101.000000	58000.000000	2.000000	3.800000
25%	102.000000	62000.000000	3.000000	3.900000
50%	103.000000	65000.000000	4.000000	4.100000

75%	104.000000	75000.000000	5.000000	4.200000
max	105.000000	78000.000000	7.000000	4.500000

----- Example 3: DataFrame Operations

Salary Statistics:

```
count      5.000000
mean      67600.000000
std       8561.541917
min       58000.000000
25%      62000.000000
50%      65000.000000
75%      75000.000000
max       78000.000000
```

Name: Salary, dtype: float64

High Performers (Score > 4.0):

	Name	Department	Performance_Score
0	Alice Johnson	Engineering	4.2
2	Charlie Brown	Engineering	4.5
4	Edward Wilson	HR	4.1

Average Salary by Department:

```
Department
Engineering    76500.0
HR              65000.0
Marketing       62000.0
Sales          58000.0
Name: Salary, dtype: float64
```

=====

3. SERIES vs DATAFRAME RELATIONSHIPS

Extracting Series from DataFrame:

Name column as Series:

```
0    Alice Johnson
1      Bob Smith
2    Charlie Brown
3      Diana Ross
4    Edward Wilson
```

Name: Name, dtype: object

Type: <class 'pandas.core.series.Series'>

----- Creating DataFrame from Multiple Series:

Product DataFrame from Series:

ID	Product	Price
----	---------	-------

```

0  201    Laptop  999.99
1  202     Mouse   25.50
2  203  Keyboard   75.00

```

```

=====
SUMMARY OF KEY DIFFERENCES:
-----

```

1. Dimensionality: Series (1D) vs DataFrame (2D)
2. Data Types: Series (homogeneous) vs DataFrame (heterogeneous per column)
3. Indexing: Series (single index) vs DataFrame (row + column indices)
4. Use Cases: Series (single variable) vs DataFrame (multiple variables)
5. Memory: Series (lower overhead) vs DataFrame (higher overhead)

Memory Usage Comparison:
 Temperature Series: 316 bytes
 Employee DataFrame: 878 bytes

1.3.5 Q2. What is re-indexing in Pandas? Explain with the example.

Answer:

Re-indexing is a fundamental operation in pandas that involves changing the row or column labels (indices) of a DataFrame or Series. It is a powerful method for data alignment, restructuring, and ensuring consistent indices across different datasets.

1.4 Concept and Mathematical Foundation

Re-indexing can be mathematically represented as a mapping function that transforms one index set to another:

Let $\mathbf{I_1} = \{i_1, i_2, \dots, i_n\}$ be the original index set
 Let $\mathbf{I_2} = \{j_1, j_2, \dots, j\}$ be the new desired index set
 Let $\mathbf{D_1}$ be the original data structure with index $\mathbf{I_1}$

The re-indexing operation creates a new data structure $\mathbf{D_2}$ with index $\mathbf{I_2}$ such that:

```

D_2[j] = {
    D_1[j]  if j  I_1 (index exists in original)
    NaN     if j  I_1 (index does not exist in original)
}

```

1.5 Core Capabilities of Re-indexing

1.5.1 1. Data Rearrangement

Re-arrange existing data according to a new sequence of labels without changing the underlying values.

1.5.2 2. Index Expansion

Add new index labels that were not present in the original dataset. Missing values are filled with NaN by default.

1.5.3 3. Index Contraction

Remove existing labels by specifying a subset of the original indices.

1.5.4 4. Data Alignment

Ensure that multiple datasets have consistent indices for mathematical operations and joins.

1.5.5 5. Time Series Resampling

Particularly useful for time series data to change frequency or fill missing time periods.

1.6 Method Signature and Parameters

The `reindex()` method has the following signature:

```
DataFrame.reindex(labels=None, index=None, columns=None, axis=None,  
                  method=None, copy=True, level=None, fill_value=NaN, limit=None)
```

Key Parameters: - **labels/index**: New row indices - **columns**: New column indices
- **method**: Method for filling missing values ('fill', 'bfill', 'nearest') - **fill_value**: Value to use for missing entries (default: NaN) - **copy**: Return a copy even if the new index is the same as the original

1.7 Practical Applications

1. **Data Preprocessing**: Standardizing indices across multiple datasets
2. **Time Series Analysis**: Creating regular time intervals
3. **Data Cleaning**: Handling missing observations
4. **Database Operations**: Simulating SQL-like joins and merges
5. **Statistical Analysis**: Ensuring proper alignment for calculations

The following examples demonstrate various re-indexing scenarios with detailed mathematical explanations:

```
[19]: # Comprehensive Re-indexing Demonstration with Mathematical Analysis  
  
print("PANDAS RE-INDEXING: COMPREHENSIVE DEMONSTRATION")  
print("=" * 60)  
  
# =====  
# PART 1: BASIC RE-INDEXING CONCEPTS  
# =====  
  
print("\n1. BASIC RE-INDEXING OPERATIONS")  
print("-" * 40)  
  
# Create original DataFrame with explicit structure  
print("Step 1: Creating Original DataFrame")  
original_data = {  
    'Sales': [1000, 1500, 1200],
```

```

        'Profit': [200, 300, 240],
        'Expenses': [800, 1200, 960]
    }
    original_index = ['Q1', 'Q2', 'Q3']

    df_original = pd.DataFrame(original_data, index=original_index)

    print("Original DataFrame D_1:")
    print(df_original)
    print(f"Original Index I_1: {list(df_original.index)}")
    print(f"Original Columns: {list(df_original.columns)}")
    print(f"Shape: {df_original.shape}")

    print("\n" + "-" * 50)

    # =====
    # PART 2: ROW RE-INDEXING EXAMPLES
    # =====

    print("\n2. ROW RE-INDEXING SCENARIOS")
    print("-" * 40)

    # Scenario A: Reordering existing indices
    print("Scenario A: Reordering Existing Indices")
    new_order = ['Q3', 'Q1', 'Q2']
    df_reordered = df_original.reindex(new_order)

    print(f"New Index Order I_2: {new_order}")
    print("Reordered DataFrame D_2:")
    print(df_reordered)

    # Mathematical verification
    print("\nMathematical Verification:")
    for old_idx, new_idx in zip(original_index, new_order):
        print(f"    D_2[{new_idx}] = D_1[{new_idx}] = {df_original.loc[new_idx, 'Sales']} (Sales)")

    print("\n" + "-" * 30)

    # Scenario B: Adding new indices (expansion)
    print("Scenario B: Index Expansion (Adding Q4)")
    expanded_index = ['Q1', 'Q2', 'Q3', 'Q4']
    df_expanded = df_original.reindex(expanded_index)

    print(f"Expanded Index I: {expanded_index}")
    print("Expanded DataFrame D:")
    print(df_expanded)

```

```

print("\nMathematical Analysis:")
print("For existing indices: D[Q_i] = D_1[Q_i] where Q_i ∈ I_1")
print("For new indices: D[Q4] = NaN (since Q4 ∉ I_1)")
print(f"Missing values count: {df_expanded.isna().sum().sum()}")

print("\n" + "-" * 30)

# Scenario C: Partial index selection (contraction)
print("Scenario C: Index Contraction (Selecting Subset)")
subset_index = ['Q1', 'Q3']
df_subset = df_original.reindex(subset_index)

print(f"Subset Index I: {subset_index}")
print("Subset DataFrame D:")
print(df_subset)

print(f"Original size: {df_original.shape[0]} rows")
print(f"Subset size: {df_subset.shape[0]} rows")
print(f>Data reduction: {((df_original.shape[0] - df_subset.shape[0]) /
→df_original.shape[0]*100):.1f}%")

print("\n" + "=" * 60)

# =====
# PART 3: COLUMN RE-INDEXING EXAMPLES
# =====

print("\n3. COLUMN RE-INDEXING SCENARIOS")
print("-" * 40)

# Scenario A: Reordering columns
print("Scenario A: Column Reordering")
new_col_order = ['Expenses', 'Sales', 'Profit']
df_col_reorder = df_original.reindex(columns=new_col_order)

print(f"New Column Order: {new_col_order}")
print("Column-Reordered DataFrame:")
print(df_col_reorder)

print("\n" + "-" * 30)

# Scenario B: Adding new columns
print("Scenario B: Adding New Columns")
expanded_columns = ['Sales', 'Profit', 'Expenses', 'Revenue', 'Growth_Rate']
df_col_expanded = df_original.reindex(columns=expanded_columns)

```

```

print(f"Expanded Columns: {expanded_columns}")
print("Column-Expanded DataFrame:")
print(df_col_expanded)

print(f"New columns added: {set(expanded_columns) - set(df_original.columns)}")
print(f"Missing values in new columns: {df_col_expanded[['Revenue', 'Growth_Rate']].isna().sum().sum()}")

print("\n" + "=" * 60)

# =====
# PART 4: ADVANCED RE-INDEXING TECHNIQUES
# =====

print("\n4. ADVANCED RE-INDEXING TECHNIQUES")
print("-" * 40)

# Fill value specification
print("Technique A: Custom Fill Values")
df_fill_zero = df_original.reindex(['Q1', 'Q2', 'Q3', 'Q4'], fill_value=0)

print("Re-indexed with fill_value=0:")
print(df_fill_zero)

print("\nComparison of fill strategies:")
df_fill_nan = df_original.reindex(['Q1', 'Q2', 'Q3', 'Q4']) # Default NaN
print("Default (NaN):")
print(df_fill_nan.loc['Q4'])
print("Custom (Zero):")
print(df_fill_zero.loc['Q4'])

print("\n" + "-" * 30)

# Forward fill and backward fill
print("Technique B: Forward Fill and Backward Fill")

# Create a DataFrame with some missing data first
time_data = pd.DataFrame({
    'Value': [10, None, 15, None, 20]
}, index=['T1', 'T2', 'T3', 'T4', 'T5'])

print("Original time series with gaps:")
print(time_data)

# Reindex with method parameters
full_time_index = ['T1', 'T1.5', 'T2', 'T2.5', 'T3', 'T3.5', 'T4', 'T4.5', 'T5']

```

```

# Forward fill
df_ffill = time_data.reindex(full_time_index, method='ffill')
print(f"\nForward Fill Method:")
print(df_ffill)

# Backward fill
df_bfill = time_data.reindex(full_time_index, method='bfill')
print(f"\nBackward Fill Method:")
print(df_bfill)

print("\n" + "=" * 60)

# =====
# PART 5: PERFORMANCE AND MEMORY ANALYSIS
# =====

print("\n5. PERFORMANCE AND MEMORY ANALYSIS")
print("-" * 40)

# Memory usage comparison
original_memory = df_original.memory_usage(deep=True).sum()
expanded_memory = df_expanded.memory_usage(deep=True).sum()

print("Memory Usage Analysis:")
print(f"Original DataFrame: {original_memory} bytes")
print(f"Expanded DataFrame: {expanded_memory} bytes")
print(f"Memory increase: {expanded_memory - original_memory} bytes")
print(f"Memory increase percentage: {((expanded_memory/original_memory - 1) * 100):.2f}%")

# Index integrity check
print(f"\nIndex Integrity Check:")
print(f"Original index is unique: {df_original.index.is_unique}")
print(f"Expanded index is unique: {df_expanded.index.is_unique}")
print(f"Original index is monotonic: {df_original.index.is_monotonic_increasing}")

print("\n" + "=" * 60)

# =====
# PART 6: PRACTICAL USE CASES
# =====

print("\n6. PRACTICAL USE CASES SUMMARY")
print("-" * 40)

print("Real-world applications of re-indexing:")

```



```

print("\n1. TIME SERIES ALIGNMENT:")
print("    - Standardizing different time frequencies")
print("    - Filling missing time periods")
print("    - Synchronizing multiple time series")

print("\n2. DATA INTEGRATION:")
print("    - Merging datasets with different indices")
print("    - Standardizing column orders across files")
print("    - Handling missing categories in categorical data")

print("\n3. STATISTICAL ANALYSIS:")
print("    - Ensuring proper alignment for calculations")
print("    - Creating balanced panels for analysis")
print("    - Preparing data for matrix operations")

print("\n4. DATABASE SIMULATION:")
print("    - Implementing LEFT/RIGHT JOIN operations")
print("    - Creating outer joins with missing data handling")
print("    - Standardizing database query results")

print("\nMathematical Properties:")
print("- Re-indexing preserves data integrity where indices match")
print("- Operation is reversible if no data is lost")
print("- Memory complexity:  $O(n \times m)$  where  $n$ =rows,  $m$ =columns")
print("- Time complexity:  $O(n)$  for index lookup and assignment")

```

PANDAS RE-INDEXING: COMPREHENSIVE DEMONSTRATION

=====

1. BASIC RE-INDEXING OPERATIONS

Step 1: Creating Original DataFrame

Original DataFrame D:

	Sales	Profit	Expenses
Q1	1000	200	800
Q2	1500	300	1200
Q3	1200	240	960

Original Index I: ['Q1', 'Q2', 'Q3']

Original Columns: ['Sales', 'Profit', 'Expenses']

Shape: (3, 3)

2. ROW RE-INDEXING SCENARIOS

Scenario A: Reordering Existing Indices

New Index Order I: ['Q3', 'Q1', 'Q2']

Reordered DataFrame D:

	Sales	Profit	Expenses
Q3	1200	240	960
Q1	1000	200	800
Q2	1500	300	1200

Mathematical Verification:

D[Q3] = D[Q3] = 1200 (Sales)
D[Q1] = D[Q1] = 1000 (Sales)
D[Q2] = D[Q2] = 1500 (Sales)

Scenario B: Index Expansion (Adding Q4)
Expanded Index I: ['Q1', 'Q2', 'Q3', 'Q4']
Expanded DataFrame D:

	Sales	Profit	Expenses
Q1	1000.0	200.0	800.0
Q2	1500.0	300.0	1200.0
Q3	1200.0	240.0	960.0
Q4	NaN	NaN	NaN

Mathematical Analysis:

For existing indices: D[Q] = D[Q] where Q ∈ I
For new indices: D[Q4] = NaN (since Q4 ∉ I)
Missing values count: 3

Scenario C: Index Contraction (Selecting Subset)
Subset Index I: ['Q1', 'Q3']
Subset DataFrame D:

	Sales	Profit	Expenses
Q1	1000	200	800
Q3	1200	240	960

Original size: 3 rows

Subset size: 2 rows

Data reduction: 33.3%

3. COLUMN RE-INDEXING SCENARIOS

Scenario A: Column Reordering

New Column Order: ['Expenses', 'Sales', 'Profit']

Column-Reordered DataFrame:

	Expenses	Sales	Profit
Q1	800	1000	200
Q2	1200	1500	300
Q3	960	1200	240

```

-----
Scenario B: Adding New Columns
Expanded Columns: ['Sales', 'Profit', 'Expenses', 'Revenue', 'Growth_Rate']
Column-Expanded DataFrame:
   Sales  Profit  Expenses  Revenue  Growth_Rate
Q1   1000    200     800     NaN      NaN
Q2   1500    300    1200     NaN      NaN
Q3   1200    240     960     NaN      NaN
New columns added: {'Revenue', 'Growth_Rate'}
Missing values in new columns: 6

```

4. ADVANCED RE-INDEXING TECHNIQUES

Technique A: Custom Fill Values

Re-indexed with fill_value=0:

	Sales	Profit	Expenses
Q1	1000	200	800
Q2	1500	300	1200
Q3	1200	240	960
Q4	0	0	0

Comparison of fill strategies:

Default (NaN):

Sales NaN

Profit NaN

Expenses NaN

Name: Q4, dtype: float64

Custom (Zero):

Sales 0

Profit 0

Expenses 0

Name: Q4, dtype: int64

Technique B: Forward Fill and Backward Fill

Original time series with gaps:

	Value
T1	10.0
T2	NaN
T3	15.0
T4	NaN
T5	20.0

Forward Fill Method:

	Value
T1	10.0

T1.5	10.0
T2	NaN
T2.5	NaN
T3	15.0
T3.5	15.0
T4	NaN
T4.5	NaN
T5	20.0

Backward Fill Method:

	Value
T1	10.0
T1.5	NaN
T2	NaN
T2.5	15.0
T3	15.0
T3.5	NaN
T4	NaN
T4.5	20.0
T5	20.0

=====

5. PERFORMANCE AND MEMORY ANALYSIS

Memory Usage Analysis:

Original DataFrame: 333 bytes

Expanded DataFrame: 300 bytes

Memory increase: -33 bytes

Memory increase percentage: -9.91%

Index Integrity Check:

Original index is unique: True

Expanded index is unique: True

Original index is monotonic: True

=====

6. PRACTICAL USE CASES SUMMARY

Real-world applications of re-indexing:

1. TIME SERIES ALIGNMENT:

- Standardizing different time frequencies
- Filling missing time periods
- Synchronizing multiple time series

2. DATA INTEGRATION:

- Merging datasets with different indices
- Standardizing column orders across files
- Handling missing categories in categorical data

3. STATISTICAL ANALYSIS:

- Ensuring proper alignment for calculations
- Creating balanced panels for analysis
- Preparing data for matrix operations

4. DATABASE SIMULATION:

- Implementing LEFT/RIGHT JOIN operations
- Creating outer joins with missing data handling
- Standardizing database query results

Mathematical Properties:

- Re-indexing preserves data integrity where indices match
- Operation is reversible if no data is lost
- Memory complexity: $O(n \times m)$ where n =rows, m =columns
- Time complexity: $O(n)$ for index lookup and assignment

1.7.1 Q3. Take a sample data and provide with the output how to compute the mean, median, standard deviation, quantile in pandas?

Answer:

Statistical measures are fundamental tools for understanding and summarizing data distributions. These measures provide insights into central tendency, dispersion, and distribution characteristics. This section demonstrates the computation of four essential statistical measures using pandas, along with their mathematical foundations and interpretations.

1.8 Theoretical Foundation

1.8.1 1. Mean (Arithmetic Average)

The mean represents the central tendency of a dataset and is calculated as the sum of all values divided by the number of observations.

Mathematical Formula:

$$\mu = (\sum_{i=1}^n x_i) / n$$

Where: - μ (mu) = population mean - x_i = individual values - n = total number of observations
 - Σ = summation symbol

Pandas Implementation: `Series.mean()` or `DataFrame.mean()`

1.8.2 2. Median (Middle Value)

The median is the middle value when data is arranged in ascending order. It is less sensitive to outliers than the mean.

Mathematical Definition: For a sorted dataset of size n :

```
Median = {
    x_n_1/_2          if n is odd
    (x_n/_2 + x_n/_2_1)/2  if n is even
}
```

Pandas Implementation: `Series.median()` or `DataFrame.median()`

1.8.3 3. Standard Deviation (Measure of Spread)

Standard deviation quantifies the amount of variation or dispersion in a dataset relative to the mean.

Mathematical Formulas:

Population Standard Deviation:

$$\sigma = [(\sum_{i=1}^n (x_i - \mu)^2) / n]$$

Sample Standard Deviation:

$$s = [(\sum_{i=1}^n (x_i - \bar{x})^2) / (n-1)]$$

Where: - σ (sigma) = population standard deviation - s = sample standard deviation
 - $\bar{x}$ = sample mean - $(n-1)$ = degrees of freedom (Bessel's correction)

Pandas Implementation: `Series.std()` (default uses sample std dev with `ddof=1`)

1.8.4 4. Quantiles (Percentiles)

Quantiles divide a dataset into equal-sized intervals. Common quantiles include quartiles (25%, 50%, 75%) and percentiles.

Mathematical Definition: For the p th percentile (where $0 \leq p \leq 1$): 1. Sort the data in ascending order 2. Calculate position: $k = p \times (n-1)$ 3. If k is integer: percentile = x_k 4. If k is not integer: interpolate between x_k and x_{k+1}

Pandas Implementation: `Series.quantile(q)` where q is between 0 and 1

1.9 Practical Importance

1. **Mean:** Useful for symmetric distributions, sensitive to outliers
2. **Median:** Robust to outliers, better for skewed distributions
3. **Standard Deviation:** Indicates data volatility and consistency
4. **Quantiles:** Provide distribution shape and identify outliers

The following comprehensive example demonstrates these calculations with detailed mathematical verification:

```
[20]: # Comprehensive Statistical Measures Computation and Analysis

print("STATISTICAL MEASURES: COMPREHENSIVE COMPUTATION AND ANALYSIS")
print("=" * 70)
```

```

# =====
# PART 1: DATASET CREATION AND EXPLORATION
# =====

print("\n1. DATASET CREATION AND INITIAL EXPLORATION")
print("-" * 50)

# Create a comprehensive student performance dataset
import numpy as np

# Set random seed for reproducibility
np.random.seed(42)

# Generate realistic student data
student_data = {
    'Student_ID': [f'STU{i:03d}' for i in range(1, 21)],
    'Mathematics': [85, 92, 78, 96, 87, 91, 83, 89, 94, 88,
                    76, 95, 82, 90, 86, 93, 79, 84, 97, 81],
    'Science': [88, 85, 92, 89, 86, 93, 87, 90, 95, 84,
                91, 83, 94, 88, 92, 86, 89, 87, 96, 85],
    'English': [76, 89, 82, 91, 85, 88, 79, 86, 93, 87,
                84, 92, 80, 94, 88, 90, 83, 85, 95, 89],
    'Physics': [90, 87, 85, 93, 89, 92, 86, 91, 96, 88,
                84, 94, 87, 91, 90, 95, 85, 89, 98, 86]
}

# Create DataFrame
scores_df = pd.DataFrame(student_data)

print("Student Performance Dataset:")
print(scores_df.head(10))
print(f"\nDataset Shape: {scores_df.shape}")
print(f"Number of Students: {scores_df.shape[0]}")
print(f"Number of Subjects: {scores_df.shape[1] - 1}") # Excluding Student_ID

# Extract numerical columns for analysis
numerical_columns = ['Mathematics', 'Science', 'English', 'Physics']
scores_only = scores_df[numerical_columns]

print(f"\nNumerical Data Overview:")
print(scores_only.head())

print("\n" + "=" * 70)

# =====
# PART 2: MEAN CALCULATION AND ANALYSIS
# =====

```

```

print("\n2. MEAN (ARITHMETIC AVERAGE) - DETAILED CALCULATION")
print("-" * 50)

print("2.1 Pandas Built-in Mean Calculation")
print("-" * 30)

# Calculate means using pandas
subject_means = scores_only.mean()
print("Subject-wise Mean Scores:")
for subject, mean_score in subject_means.items():
    print(f" {subject}: {mean_score:.4f}")

overall_mean = scores_only.mean().mean()
print(f"\nOverall Mean (across all subjects): {overall_mean:.4f}")

print("\n2.2 Manual Mean Calculation with Mathematical Verification")
print("-" * 30)

# Manual calculation for Mathematics
math_scores = scores_only['Mathematics'].values
n_students = len(math_scores)

print(f"Mathematics Scores: {math_scores}")
print(f"Number of students (n): {n_students}")

# Step-by-step calculation
sum_scores = sum(math_scores)
manual_mean = sum_scores / n_students

print(f"\nStep-by-step calculation:")
print(f" $\Sigma x_i = \{ ' + '.join(map(str, math_scores)) \} = \{sum\_scores\}$ ")
print(f" $\mu = \Sigma x_i / n = \{sum\_scores\} / \{n\_students\} = \{manual\_mean:.4f\}$ ")
print(f"Pandas mean(): {scores_only['Mathematics'].mean():.4f}")
print(f"Verification: Manual = Pandas? {abs(manual_mean -
    ↳ scores_only['Mathematics'].mean()) < 1e-10}")

print("\n2.3 Statistical Interpretation")
print("-" * 30)
print("The mean represents the expected value of the distribution.")
print(f"On average, students scored {manual_mean:.1f} points in Mathematics.")
print(f"This suggests {'above average' if manual_mean > 85 else 'average' if
    ↳ manual_mean >= 80 else 'below average'} performance.")

print("\n" + "=" * 70)

# =====

```



```

# PART 3: MEDIAN CALCULATION AND ANALYSIS
# =====

print("\n3. MEDIAN (MIDDLE VALUE) - DETAILED CALCULATION")
print("-" * 50)

print("3.1 Pandas Built-in Median Calculation")
print("-" * 30)

subject_medians = scores_only.median()
print("Subject-wise Median Scores:")
for subject, median_score in subject_medians.items():
    print(f" {subject}: {median_score:.4f}")

print("\n3.2 Manual Median Calculation with Mathematical Verification")
print("-" * 30)

# Manual calculation for Mathematics
math_sorted = sorted(math_scores)
print(f"Sorted Mathematics Scores: {math_sorted}")
print(f"Number of values (n): {n_students}")

if n_students % 2 == 1:
    # Odd number of values
    middle_index = n_students // 2
    manual_median = math_sorted[middle_index]
    print(f"\nOdd number of values:")
    print(f"Middle position:  $(\{n\_students\}+1)/2 = \{(n\_students+1)//2\}$ ")
    print(f"Median =  $x_{\{middle\_index+1\}} = \{manual\_median\}$ ")
else:
    # Even number of values
    middle1 = n_students // 2 - 1
    middle2 = n_students // 2
    manual_median = (math_sorted[middle1] + math_sorted[middle2]) / 2
    print(f"\nEven number of values:")
    print(f"Middle positions:  $\{middle1+1\}$  and  $\{middle2+1\}$ ")
    print(f"Values:  $\{math\_sorted[middle1]\}$  and  $\{math\_sorted[middle2]\}$ ")
    print(f"Median =  $(\{math\_sorted[middle1]\} + \{math\_sorted[middle2]\})/2 = \{manual\_median\}$ ")

print(f"Pandas median(): {scores_only['Mathematics'].median():.4f}")
print(f"Verification: Manual = Pandas?  $\{abs(manual\_median - \rightarrow scores\_only['Mathematics'].median()) < 1e-10\}$ ")

print("\n3.3 Mean vs Median Comparison")
print("-" * 30)
for subject in numerical_columns:

```

```

    mean_val = scores_only[subject].mean()
    median_val = scores_only[subject].median()
    skew_indicator = "Right-skewed" if mean_val > median_val else "Left-skewed"
    → if mean_val < median_val else "Symmetric"
    print(f"{subject}:")
    print(f"    Mean: {mean_val:.2f}, Median: {median_val:.2f}")
    print(f"    Distribution: {skew_indicator}")

print("\n" + "=" * 70)

# =====
# PART 4: STANDARD DEVIATION CALCULATION
# =====

print("\n4. STANDARD DEVIATION - DETAILED CALCULATION")
print("-" * 50)

print("4.1 Pandas Built-in Standard Deviation")
print("-" * 30)

subject_std = scores_only.std() # Sample standard deviation (ddof=1)
subject_std_pop = scores_only.std(ddof=0) # Population standard deviation

print("Sample Standard Deviation (ddof=1):")
for subject, std_score in subject_std.items():
    print(f"    {subject}: {std_score:.4f}")

print("\nPopulation Standard Deviation (ddof=0):")
for subject, std_score in subject_std_pop.items():
    print(f"    {subject}: {std_score:.4f}")

print("\n4.2 Manual Standard Deviation Calculation")
print("-" * 30)

# Manual calculation for Mathematics (Sample Standard Deviation)
math_mean = scores_only['Mathematics'].mean()
print(f"Mathematics scores: {math_scores}")
print(f"Mean (x): {math_mean:.4f}")

# Calculate deviations from mean
deviations = math_scores - math_mean
print(f"\nDeviations from mean (x_i - x):")
for i, (score, dev) in enumerate(zip(math_scores, deviations)):
    print(f"    Student {i+1}: {score} - {math_mean:.4f} = {dev:.4f}")

# Calculate squared deviations
squared_deviations = deviations ** 2

```

```

print(f"\nSquared deviations (x_i - x)²:")
for i, (dev, sq_dev) in enumerate(zip(deviations, squared_deviations)):
    print(f"  Student {i+1}: ({dev:.4f})² = {sq_dev:.4f}")

# Sample variance and standard deviation
sum_squared_dev = sum(squared_deviations)
sample_variance = sum_squared_dev / (n_students - 1) # Bessel's correction
sample_std = np.sqrt(sample_variance)

print(f"\nSample Variance Calculation:")
print(f"Σ(x_i - x)² = {sum_squared_dev:.4f}")
print(f"s² = Σ(x_i - x)²/(n-1) = {sum_squared_dev:.4f}/{n_students-1} = □
    ↪{sample_variance:.4f}")
print(f"s = (s²) = {sample_variance:.4f} = {sample_std:.4f}")

print(f"\nPandas std(): {scores_only['Mathematics'].std():.4f}")
print(f"Verification: Manual = Pandas? {abs(sample_std - □
    ↪scores_only['Mathematics'].std()) < 1e-10}")

print("\n4.3 Standard Deviation Interpretation")
print("-" * 30)
for subject in numerical_columns:
    mean_val = scores_only[subject].mean()
    std_val = scores_only[subject].std()
    cv = (std_val / mean_val) * 100 # Coefficient of variation

    print(f"{subject}:")
    print(f"  Mean: {mean_val:.2f}, Std Dev: {std_val:.2f}")
    print(f"  Coefficient of Variation: {cv:.2f}%")
    print(f"  Interpretation: {'Low variability' if cv < 10 else 'Moderate'
    ↪variability' if cv < 20 else 'High variability'}")

print("\n" + "=" * 70)

# =====
# PART 5: QUANTILES CALCULATION AND ANALYSIS
# =====

print("\n5. QUANTILES (PERCENTILES) - DETAILED CALCULATION")
print("-" * 50)

print("5.1 Pandas Built-in Quantile Calculation")
print("-" * 30)

# Common quantiles
quantiles_to_calc = [0.25, 0.5, 0.75]

```

```

quantile_names = ['Q1 (25th percentile)', 'Q2 (50th percentile/Median)', 'Q3 (75th percentile)']

print("Subject-wise Quantiles:")
for subject in numerical_columns:
    print(f"\n{subject}:")
    for q, name in zip(quantiles_to_calc, quantile_names):
        q_value = scores_only[subject].quantile(q)
        print(f"    {name}: {q_value:.4f}")

print("\n5.2 Manual Quantile Calculation")
print("-" * 30)

# Manual calculation for Mathematics Q1 (25th percentile)
print(f"Mathematics scores (sorted): {math_sorted}")
print(f"Number of values (n): {n_students}")

# Calculate Q1 position
q1_position = 0.25 * (n_students - 1)
print(f"\nQ1 Position Calculation:")
print(f"Position = p × (n-1) = 0.25 × ({n_students}-1) = 0.25 × {n_students-1} = {q1_position}")

if q1_position == int(q1_position):
    # Position is integer
    q1_manual = math_sorted[int(q1_position)]
    print(f"Position is integer: Q1 = x[{int(q1_position)+1}] = {q1_manual}")
else:
    # Interpolation required
    lower_index = int(q1_position)
    upper_index = lower_index + 1
    fraction = q1_position - lower_index

    q1_manual = math_sorted[lower_index] + fraction * (math_sorted[upper_index] - math_sorted[lower_index])
    print(f"Position requires interpolation:")
    print(f"Lower index: {lower_index+1}, Value: {math_sorted[lower_index]}")
    print(f"Upper index: {upper_index+1}, Value: {math_sorted[upper_index]}")
    print(f"Fraction: {fraction:.4f}")
    print(f"Q1 = {math_sorted[lower_index]} + {fraction:.4f} × ({math_sorted[upper_index]} - {math_sorted[lower_index]}) = {q1_manual:.4f}")

pandas_q1 = scores_only['Mathematics'].quantile(0.25)
print(f"\nPandas quantile(0.25): {pandas_q1:.4f}")
print(f"Verification: |Manual - Pandas| = {abs(q1_manual - pandas_q1):.4f}")

print("\n5.3 Quartile Analysis and Interpretation")

```

```

print("-" * 30)

for subject in numerical_columns:
    q1 = scores_only[subject].quantile(0.25)
    q2 = scores_only[subject].quantile(0.5) # Median
    q3 = scores_only[subject].quantile(0.75)
    iqr = q3 - q1 # Interquartile Range

    print(f"\n{subject} Quartile Analysis:")
    print(f"  Q1 (25%): {q1:.2f} - 25% of students scored below this")
    print(f"  Q2 (50%): {q2:.2f} - Median score")
    print(f"  Q3 (75%): {q3:.2f} - 75% of students scored below this")
    print(f"  IQR: {iqr:.2f} - Middle 50% of scores span this range")

    # Outlier detection using IQR method
    lower_fence = q1 - 1.5 * iqr
    upper_fence = q3 + 1.5 * iqr
    outliers = scores_only[subject][(scores_only[subject] < lower_fence) |
    ↪(scores_only[subject] > upper_fence)]

    print(f"  Outlier boundaries: [{lower_fence:.2f}, {upper_fence:.2f}]")
    print(f"  Outliers detected: {len(outliers)} ({list(outliers.values) if
    ↪len(outliers) > 0 else 'None'})")

print("\n" + "=" * 70)

# =====
# PART 6: COMPREHENSIVE STATISTICAL SUMMARY
# =====

print("\n6. COMPREHENSIVE STATISTICAL SUMMARY")
print("-" * 50)

# Generate comprehensive summary
summary_stats = scores_only.describe()
print("Complete Statistical Summary:")
print(summary_stats)

print("\n6.1 Custom Summary Statistics")
print("-" * 30)

custom_summary = pd.DataFrame({
    'Count': scores_only.count(),
    'Mean': scores_only.mean(),
    'Median': scores_only.median(),
    'Mode': scores_only.mode().iloc[0], # First mode if multiple
    'Std_Dev': scores_only.std(),

```

```

    'Variance': scores_only.var(),
    'Skewness': scores_only.skew(),
    'Kurtosis': scores_only.kurtosis(),
    'Range': scores_only.max() - scores_only.min(),
    'IQR': scores_only.quantile(0.75) - scores_only.quantile(0.25)
})

print(custom_summary.round(4))

print("\n6.2 Key Statistical Insights")
print("-" * 30)

for subject in numerical_columns:
    mean_val = scores_only[subject].mean()
    median_val = scores_only[subject].median()
    std_val = scores_only[subject].std()
    skew_val = scores_only[subject].skew()

    print(f"\n{subject} Insights:")
    print(f"  Central Tendency: Mean={mean_val:.1f}, Median={median_val:.1f}")
    print(f"  Variability: Std Dev={std_val:.2f} ({'Low' if std_val < 4 else 'Moderate' if std_val < 6 else 'High'} variation)")
    print(f"  Distribution Shape: {'Symmetric' if abs(skew_val) < 0.5 else 'Moderately Skewed' if abs(skew_val) < 1 else 'Highly Skewed'}")
    print(f"  Performance Level: {'Excellent' if mean_val >= 90 else 'Good' if mean_val >= 85 else 'Average' if mean_val >= 80 else 'Below Average'}")

print("\n" + "=" * 70)
print("STATISTICAL COMPUTATION COMPLETED SUCCESSFULLY")
print("All calculations verified against pandas built-in functions")
print("=" * 70)

```

STATISTICAL MEASURES: COMPREHENSIVE COMPUTATION AND ANALYSIS

=====

1. DATASET CREATION AND INITIAL EXPLORATION

Student Performance Dataset:

	Student_ID	Mathematics	Science	English	Physics
0	STU001	85	88	76	90
1	STU002	92	85	89	87
2	STU003	78	92	82	85
3	STU004	96	89	91	93
4	STU005	87	86	85	89
5	STU006	91	93	88	92
6	STU007	83	87	79	86
7	STU008	89	90	86	91

8	STU009	94	95	93	96
9	STU010	88	84	87	88

Dataset Shape: (20, 5)
 Number of Students: 20
 Number of Subjects: 4

Numerical Data Overview:

	Mathematics	Science	English	Physics
0	85	88	76	90
1	92	85	89	87
2	78	92	82	85
3	96	89	91	93
4	87	86	85	89

=====

2. MEAN (ARITHMETIC AVERAGE) - DETAILED CALCULATION

2.1 Pandas Built-in Mean Calculation

Subject-wise Mean Scores:

Mathematics: 87.3000
 Science: 89.0000
 English: 86.8000
 Physics: 89.8000

Overall Mean (across all subjects): 88.2250

2.2 Manual Mean Calculation with Mathematical Verification

Mathematics Scores: [85 92 78 96 87 91 83 89 94 88 76 95 82 90 86 93 79 84 97 81]

Number of students (n): 20

Step-by-step calculation:

$\Sigma x = 85 + 92 + 78 + 96 + 87 + 91 + 83 + 89 + 94 + 88 + 76 + 95 + 82 + 90 + 86 + 93 + 79 + 84 + 97 + 81 = 1746$

$\mu = \Sigma x / n = 1746 / 20 = 87.3000$

Pandas mean(): 87.3000

Verification: Manual = Pandas? True

2.3 Statistical Interpretation

The mean represents the expected value of the distribution.
 On average, students scored 87.3 points in Mathematics.
 This suggests above average performance.

=====

3. MEDIAN (MIDDLE VALUE) - DETAILED CALCULATION

3.1 Pandas Built-in Median Calculation

Subject-wise Median Scores:

Mathematics: 87.5000

Science: 88.5000

English: 87.5000

Physics: 89.5000

3.2 Manual Median Calculation with Mathematical Verification

Sorted Mathematics Scores: [np.int64(76), np.int64(78), np.int64(79),
np.int64(81), np.int64(82), np.int64(83), np.int64(84), np.int64(85),
np.int64(86), np.int64(87), np.int64(88), np.int64(89), np.int64(90),
np.int64(91), np.int64(92), np.int64(93), np.int64(94), np.int64(95),
np.int64(96), np.int64(97)]

Number of values (n): 20

Even number of values:

Middle positions: 10 and 11

Values: 87 and 88

Median = $(87 + 88)/2 = 87.5$

Pandas median(): 87.5000

Verification: Manual = Pandas? True

3.3 Mean vs Median Comparison

Mathematics:

Mean: 87.30, Median: 87.50

Distribution: Left-skewed

Science:

Mean: 89.00, Median: 88.50

Distribution: Right-skewed

English:

Mean: 86.80, Median: 87.50

Distribution: Left-skewed

Physics:

Mean: 89.80, Median: 89.50

Distribution: Right-skewed

=====

4. STANDARD DEVIATION - DETAILED CALCULATION

4.1 Pandas Built-in Standard Deviation

Sample Standard Deviation (ddof=1):

Mathematics: 6.2416
Science: 3.7697
English: 5.1360
Physics: 3.9550

Population Standard Deviation (ddof=0):

Mathematics: 6.0836
Science: 3.6742
English: 5.0060
Physics: 3.8549

4.2 Manual Standard Deviation Calculation

Mathematics scores: [85 92 78 96 87 91 83 89 94 88 76 95 82 90 86 93 79 84 97 81]

Mean (x): 87.3000

Deviations from mean (x - x):

Student 1: 85 - 87.3000 = -2.3000
Student 2: 92 - 87.3000 = 4.7000
Student 3: 78 - 87.3000 = -9.3000
Student 4: 96 - 87.3000 = 8.7000
Student 5: 87 - 87.3000 = -0.3000
Student 6: 91 - 87.3000 = 3.7000
Student 7: 83 - 87.3000 = -4.3000
Student 8: 89 - 87.3000 = 1.7000
Student 9: 94 - 87.3000 = 6.7000
Student 10: 88 - 87.3000 = 0.7000
Student 11: 76 - 87.3000 = -11.3000
Student 12: 95 - 87.3000 = 7.7000
Student 13: 82 - 87.3000 = -5.3000
Student 14: 90 - 87.3000 = 2.7000
Student 15: 86 - 87.3000 = -1.3000
Student 16: 93 - 87.3000 = 5.7000
Student 17: 79 - 87.3000 = -8.3000
Student 18: 84 - 87.3000 = -3.3000
Student 19: 97 - 87.3000 = 9.7000
Student 20: 81 - 87.3000 = -6.3000

Squared deviations (x - x)²:

Student 1: (-2.3000)² = 5.2900
Student 2: (4.7000)² = 22.0900
Student 3: (-9.3000)² = 86.4900
Student 4: (8.7000)² = 75.6900
Student 5: (-0.3000)² = 0.0900
Student 6: (3.7000)² = 13.6900

Student 7: $(-4.3000)^2 = 18.4900$
 Student 8: $(1.7000)^2 = 2.8900$
 Student 9: $(6.7000)^2 = 44.8900$
 Student 10: $(0.7000)^2 = 0.4900$
 Student 11: $(-11.3000)^2 = 127.6900$
 Student 12: $(7.7000)^2 = 59.2900$
 Student 13: $(-5.3000)^2 = 28.0900$
 Student 14: $(2.7000)^2 = 7.2900$
 Student 15: $(-1.3000)^2 = 1.6900$
 Student 16: $(5.7000)^2 = 32.4900$
 Student 17: $(-8.3000)^2 = 68.8900$
 Student 18: $(-3.3000)^2 = 10.8900$
 Student 19: $(9.7000)^2 = 94.0900$
 Student 20: $(-6.3000)^2 = 39.6900$

Sample Variance Calculation:

$\Sigma(x - \bar{x})^2 = 740.2000$

$s^2 = \Sigma(x - \bar{x})^2 / (n-1) = 740.2000 / 19 = 38.9579$

$s = (s^2)^{1/2} = 38.9579 = 6.2416$

Pandas std(): 6.2416

Verification: Manual = Pandas? True

4.3 Standard Deviation Interpretation

Mathematics:

Mean: 87.30, Std Dev: 6.24

Coefficient of Variation: 7.15%

Interpretation: Low variability

Science:

Mean: 89.00, Std Dev: 3.77

Coefficient of Variation: 4.24%

Interpretation: Low variability

English:

Mean: 86.80, Std Dev: 5.14

Coefficient of Variation: 5.92%

Interpretation: Low variability

Physics:

Mean: 89.80, Std Dev: 3.96

Coefficient of Variation: 4.40%

Interpretation: Low variability

=====

5. QUANTILES (PERCENTILES) - DETAILED CALCULATION

5.1 Pandas Built-in Quantile Calculation

Subject-wise Quantiles:

Mathematics:

Q1 (25th percentile): 82.7500
Q2 (50th percentile/Median): 87.5000
Q3 (75th percentile): 92.2500

Science:

Q1 (25th percentile): 86.0000
Q2 (50th percentile/Median): 88.5000
Q3 (75th percentile): 92.0000

English:

Q1 (25th percentile): 83.7500
Q2 (50th percentile/Median): 87.5000
Q3 (75th percentile): 90.2500

Physics:

Q1 (25th percentile): 86.7500
Q2 (50th percentile/Median): 89.5000
Q3 (75th percentile): 92.2500

5.2 Manual Quantile Calculation

Mathematics scores (sorted): [np.int64(76), np.int64(78), np.int64(79),
np.int64(81), np.int64(82), np.int64(83), np.int64(84), np.int64(85),
np.int64(86), np.int64(87), np.int64(88), np.int64(89), np.int64(90),
np.int64(91), np.int64(92), np.int64(93), np.int64(94), np.int64(95),
np.int64(96), np.int64(97)]
Number of values (n): 20

Q1 Position Calculation:

Position = $p \times (n-1) = 0.25 \times (20-1) = 0.25 \times 19 = 4.75$

Position requires interpolation:

Lower index: 5, Value: 82

Upper index: 6, Value: 83

Fraction: 0.7500

$Q1 = 82 + 0.7500 \times (83 - 82) = 82.7500$

Pandas quantile(0.25): 82.7500

Verification: |Manual - Pandas| = 0.0000

5.3 Quartile Analysis and Interpretation

Mathematics Quartile Analysis:

Q1 (25%): 82.75 - 25% of students scored below this
Q2 (50%): 87.50 - Median score

Q3 (75%): 92.25 - 75% of students scored below this
 IQR: 9.50 - Middle 50% of scores span this range
 Outlier boundaries: [68.50, 106.50]
 Outliers detected: 0 (None)

Science Quartile Analysis:

Q1 (25%): 86.00 - 25% of students scored below this
 Q2 (50%): 88.50 - Median score
 Q3 (75%): 92.00 - 75% of students scored below this
 IQR: 6.00 - Middle 50% of scores span this range
 Outlier boundaries: [77.00, 101.00]
 Outliers detected: 0 (None)

English Quartile Analysis:

Q1 (25%): 83.75 - 25% of students scored below this
 Q2 (50%): 87.50 - Median score
 Q3 (75%): 90.25 - 75% of students scored below this
 IQR: 6.50 - Middle 50% of scores span this range
 Outlier boundaries: [74.00, 100.00]
 Outliers detected: 0 (None)

Physics Quartile Analysis:

Q1 (25%): 86.75 - 25% of students scored below this
 Q2 (50%): 89.50 - Median score
 Q3 (75%): 92.25 - 75% of students scored below this
 IQR: 5.50 - Middle 50% of scores span this range
 Outlier boundaries: [78.50, 100.50]
 Outliers detected: 0 (None)

=====

6. COMPREHENSIVE STATISTICAL SUMMARY

Complete Statistical Summary:

	Mathematics	Science	English	Physics
count	20.000000	20.000000	20.000000	20.000000
mean	87.300000	89.000000	86.800000	89.800000
std	6.241626	3.769685	5.136044	3.95501
min	76.000000	83.000000	76.000000	84.000000
25%	82.750000	86.000000	83.750000	86.750000
50%	87.500000	88.500000	87.500000	89.500000
75%	92.250000	92.000000	90.250000	92.250000
max	97.000000	96.000000	95.000000	98.000000

6.1 Custom Summary Statistics

	Count	Mean	Median	Mode	Std_Dev	Variance	Skewness	Kurtosis	\
Mathematics	20	87.3	87.5	76.0	6.2416	38.9579	-0.1575	-1.0027	

Science	20	89.0	88.5	85.0	3.7697	14.2105	0.2947	-0.8970
English	20	86.8	87.5	85.0	5.1360	26.3789	-0.3510	-0.4091
Physics	20	89.8	89.5	85.0	3.9550	15.6421	0.4485	-0.6109

	Range	IQR
Mathematics	21	9.5
Science	13	6.0
English	19	6.5
Physics	14	5.5

6.2 Key Statistical Insights

Mathematics Insights:

Central Tendency: Mean=87.3, Median=87.5
 Variability: Std Dev=6.24 (High variation)
 Distribution Shape: Symmetric
 Performance Level: Good

Science Insights:

Central Tendency: Mean=89.0, Median=88.5
 Variability: Std Dev=3.77 (Low variation)
 Distribution Shape: Symmetric
 Performance Level: Good

English Insights:

Central Tendency: Mean=86.8, Median=87.5
 Variability: Std Dev=5.14 (Moderate variation)
 Distribution Shape: Symmetric
 Performance Level: Good

Physics Insights:

Central Tendency: Mean=89.8, Median=89.5
 Variability: Std Dev=3.96 (Low variation)
 Distribution Shape: Symmetric
 Performance Level: Good

```
=====
STATISTICAL COMPUTATION COMPLETED SUCCESSFULLY
All calculations verified against pandas built-in functions
=====
```

1.9.1 Q4. What is the difference between ndarray and array in Numpy? How would you convert a Pandas dataframe into a Numpy array?

Mathematical Foundation and Data Structure Theory The distinction between `ndarray` and `array` in NumPy is fundamental to understanding NumPy's architecture and data representation in scientific computing.

Theoretical Analysis 1. NumPy ndarray (N-dimensional array):

An **ndarray** is the core data structure in NumPy, representing homogeneous multidimensional arrays. Mathematically, an ndarray can be viewed as a tensor $\mathbf{T}$ of rank n :

$$\mathbf{T} \in \mathbb{R}^{d_1 \times d_2 \times \dots \times d_n}$$

Where: - d_i represents the size of the i -th dimension - n is the number of dimensions (rank of the tensor) - All elements share the same data type for memory efficiency

Memory Layout: NumPy arrays use contiguous memory allocation following either: - **C-order (row-major)**: Elements stored row by row - **Fortran-order (column-major)**: Elements stored column by column

2. np.array() Function:

The `np.array()` function is a **constructor** that creates ndarray objects. It performs several operations:

1. **Type inference:** Automatically determines the most appropriate dtype
2. **Memory allocation:** Allocates contiguous memory block
3. **Data copying:** Creates a copy of input data (unless specified otherwise)

Conversion Methods: DataFrame $\rightarrow$ NumPy Array Mathematical Representation:

Given a Pandas DataFrame $\mathbf{D}$ with shape (m, n) :

$$\mathbf{D} = \begin{pmatrix} d_{1,1} & d_{1,2} & \dots & d_{1,n} \\ d_{2,1} & d_{2,2} & \dots & d_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ d_{m,1} & d_{m,2} & \dots & d_{m,n} \end{pmatrix}$$

Conversion transforms this to NumPy array $\mathbf{A}$:

$$\mathbf{A} = \text{numpy_array}(\mathbf{D}) \in \mathbb{R}^{m \times n}$$

Available Methods:

1. **.values attribute** (Legacy):

$$\mathbf{A} = \mathbf{D}.\text{values}$$

2. **.to_numpy() method** (Recommended):

$$\mathbf{A} = \mathbf{D}.\text{to_numpy}()$$

3. **Column-specific conversion:**

$$\mathbf{a}_j = \mathbf{D}[\text{column}_j].\text{to_numpy}()$$

Performance Considerations **Time Complexity:** $O(mn)$ for copying all elements **Space Complexity:** $O(mn)$ for creating new array (unless using views)

Memory Efficiency Formula:

$$\text{Memory} = m \times n \times \text{sizeof}(\text{dtype})$$

This conversion enables leveraging NumPy's vectorized operations, which provide significant performance improvements through: - SIMD (Single Instruction, Multiple Data) operations - Cache-efficient memory access patterns - Optimized linear algebra libraries (BLAS/LAPACK)

Cross-Reference See *statistical_formulas_reference.ipynb* Section 5.1 for detailed matrix operations and *01_descriptive_statistics_interactive.ipynb* for practical array manipulation examples.

```
[21]: # Demonstration of ndarray vs array
import numpy as np

print("=" * 80)
print("Q4: NUMPY NDARRAY vs ARRAY FUNCTION DEMONSTRATION")
print("=" * 80)

# Creating an array using np.array() function
arr = np.array([1, 2, 3, 4, 5])
print("\n1. ARRAY CREATION AND TYPE ANALYSIS")
print("-" * 40)
print(f"Created using np.array(): {arr}")
print(f"Type: {type(arr)}")
print(f"Is it ndarray? {isinstance(arr, np.ndarray)}")
print(f"Data type: {arr.dtype}")
print(f"Shape: {arr.shape}")
print(f"Number of dimensions: {arr.ndim}")

print("\n" + "="*60)

# Convert Pandas DataFrame to NumPy array
print("\n2. PANDAS DATAFRAME TO NUMPY ARRAY CONVERSION")
print("-" * 50)
print("Original DataFrame (Student Performance Data):")
print(scores_df.head())
print(f"DataFrame shape: {scores_df.shape}")
print(f"DataFrame dtypes:\n{scores_df.dtypes}")

print("\n2.1 Method 1: Using .values attribute (Legacy approach)")
print("-" * 30)
numpy_array1 = scores_df.values
print(f"Type: {type(numpy_array1)}")
print(f"Shape: {numpy_array1.shape}")
print(f"Data type: {numpy_array1.dtype}")
```

```

print("First 5 rows of NumPy array:")
print(numpy_array1[:5])

print("\n2.2 Method 2: Using .to_numpy() method (Recommended)")
print("-" * 30)
numpy_array2 = scores_df.to_numpy()
print(f"Type: {type(numpy_array2)}")
print(f"Shape: {numpy_array2.shape}")
print(f"Data type: {numpy_array2.dtype}")
print("Array equality check:", np.array_equal(numpy_array1, numpy_array2))

print("\n2.3 Method 3: Converting specific numerical columns")
print("-" * 30)
numerical_data = scores_df[['Mathematics', 'Science', 'English', 'Physics']].
    ↪to_numpy()
print(f"Numerical data shape: {numerical_data.shape}")
print(f"Data type: {numerical_data.dtype}")
print("First 5 rows of numerical data:")
print(numerical_data[:5])

print("\n2.4 Method 4: Converting single column to 1D array")
print("-" * 30)
math_scores_array = scores_df['Mathematics'].to_numpy()
print(f"Math scores array: {math_scores_array}")
print(f"Type: {type(math_scores_array)}")
print(f"Shape: {math_scores_array.shape}")
print(f"Mean score: {np.mean(math_scores_array):.2f}")

print("\n" + "="*60)

print("\n3. MATHEMATICAL ANALYSIS OF CONVERSION")
print("-" * 40)

# Demonstrate mathematical operations possible after conversion
print("Original DataFrame statistics:")
print(scores_df.describe())

print("\nNumPy array statistics (numerical columns only):")
print(f"Mean values: {np.mean(numerical_data, axis=0)}")
print(f"Standard deviations: {np.std(numerical_data, axis=0, ddof=1)}")
print(f"Correlation matrix shape: {np.corrcoef(numerical_data.T).shape}")

print("\n4. MEMORY AND PERFORMANCE COMPARISON")
print("-" * 40)
import sys

df_memory = scores_df.memory_usage(deep=True).sum()

```



```

array_memory = numerical_data.nbytes

print(f"DataFrame memory usage: {df_memory} bytes")
print(f"NumPy array memory usage: {array_memory} bytes")
print(f"Memory efficiency ratio: {array_memory/df_memory:.3f}")

print("\n" + "="*80)

```

Q4: NUMPY NDARRAY vs ARRAY FUNCTION DEMONSTRATION

1. ARRAY CREATION AND TYPE ANALYSIS

```

-----
Created using np.array(): [1 2 3 4 5]
Type: <class 'numpy.ndarray'>
Is it ndarray? True
Data type: int64
Shape: (5,)
Number of dimensions: 1

```

2. PANDAS DATAFRAME TO NUMPY ARRAY CONVERSION

Original DataFrame (Student Performance Data):

	Student_ID	Mathematics	Science	English	Physics
0	STU001	85	88	76	90
1	STU002	92	85	89	87
2	STU003	78	92	82	85
3	STU004	96	89	91	93
4	STU005	87	86	85	89

DataFrame shape: (20, 5)

DataFrame dtypes:

```

Student_ID      object
Mathematics      int64
Science          int64
English          int64
Physics          int64
dtype: object

```

2.1 Method 1: Using .values attribute (Legacy approach)

```

-----
Type: <class 'numpy.ndarray'>
Shape: (20, 5)
Data type: object
First 5 rows of NumPy array:
[['STU001' 85 88 76 90]

```

```
['STU002' 92 85 89 87]
['STU003' 78 92 82 85]
['STU004' 96 89 91 93]
['STU005' 87 86 85 89]]
```

2.2 Method 2: Using .to_numpy() method (Recommended)

```
-----
Type: <class 'numpy.ndarray'>
Shape: (20, 5)
Data type: object
Array equality check: True
```

2.3 Method 3: Converting specific numerical columns

```
-----
Numerical data shape: (20, 4)
Data type: int64
First 5 rows of numerical data:
[[85 88 76 90]
 [92 85 89 87]
 [78 92 82 85]
 [96 89 91 93]
 [87 86 85 89]]
```

2.4 Method 4: Converting single column to 1D array

```
-----
Math scores array: [85 92 78 96 87 91 83 89 94 88 76 95 82 90 86 93 79 84 97 81]
Type: <class 'numpy.ndarray'>
Shape: (20,)
Mean score: 87.30
```

```
=====
```

3. MATHEMATICAL ANALYSIS OF CONVERSION

```
-----
Original DataFrame statistics:
```

	Mathematics	Science	English	Physics
count	20.000000	20.000000	20.000000	20.000000
mean	87.300000	89.000000	86.800000	89.800000
std	6.241626	3.769685	5.136044	3.95501
min	76.000000	83.000000	76.000000	84.00000
25%	82.750000	86.000000	83.750000	86.75000
50%	87.500000	88.500000	87.500000	89.50000
75%	92.250000	92.000000	90.250000	92.25000
max	97.000000	96.000000	95.000000	98.00000

```
NumPy array statistics (numerical columns only):
Mean values: [87.3 89. 86.8 89.8]
Standard deviations: [6.24162597 3.76968517 5.13604394 3.95501015]
```

Correlation matrix shape: (4, 4)

4. MEMORY AND PERFORMANCE COMPARISON

DataFrame memory usage: 1872 bytes

NumPy array memory usage: 640 bytes

Memory efficiency ratio: 0.342

=====

1.9.2 Q5. What is the difference between `test[:, 0]` and `test[:, [0]]`? Explain with example the difference between Vectorisation and Broadcasting in NumPy?

Mathematical Foundation of Array Indexing and Operations Understanding array indexing, vectorization, and broadcasting is crucial for efficient numerical computing and forms the basis of modern data science operations.

Theoretical Analysis of Array Indexing Given a matrix $\mathbf{T} \in \mathbb{R}^{m \times n}$:

$$\mathbf{T} = \begin{pmatrix} t_{1,1} & t_{1,2} & \cdots & t_{1,n} \\ t_{2,1} & t_{2,2} & \cdots & t_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ t_{m,1} & t_{m,2} & \cdots & t_{m,n} \end{pmatrix}$$

1. `test[:, 0]` - Fancy Indexing (Returns 1D array):

This operation extracts the first column as a 1D vector:

$$\mathbf{v} = \mathbf{T}[:, 0] = \begin{pmatrix} t_{1,1} \\ t_{2,1} \\ \vdots \\ t_{m,1} \end{pmatrix} \in \mathbb{R}^m$$

Mathematical properties: - **Shape:** $(m,)$ - 1-dimensional - **Rank:** 1 (vector) - **Memory layout:** Contiguous 1D array

2. `test[:, [0]]` - Advanced Indexing (Returns 2D array):

This operation extracts the first column while preserving 2D structure:

$$\mathbf{M} = \mathbf{T}[:, [0]] = \begin{pmatrix} t_{1,1} \\ t_{2,1} \\ \vdots \\ t_{m,1} \end{pmatrix} \in \mathbb{R}^{m \times 1}$$

Mathematical properties: - **Shape:** $(m, 1)$ - 2-dimensional - **Rank:** 2 (matrix) - **Memory layout:** 2D array with single column

Vectorization Theory Definition: Vectorization is the process of applying operations to entire arrays without explicit Python loops, leveraging optimized C/Fortran implementations.

Mathematical Representation:

Instead of element-wise operations:

$$\forall i : c_i = f(a_i, b_i)$$

Vectorization performs:

$$\mathbf{c} = f(\mathbf{a}, \mathbf{b})$$

Performance Analysis: - **Time Complexity:** $O(n)$ with optimized constants - **Memory Access:** Sequential, cache-friendly - **SIMD Utilization:** Multiple operations per CPU instruction

Example Operation:

$$\mathbf{a} \odot \mathbf{b} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \odot \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} = \begin{pmatrix} a_1 \cdot b_1 \\ a_2 \cdot b_2 \\ \vdots \\ a_n \cdot b_n \end{pmatrix}$$

Broadcasting Theory Definition: Broadcasting allows NumPy to perform arithmetic operations on arrays with different shapes by automatically expanding dimensions according to specific rules.

Broadcasting Rules:

1. **Rule 1:** Arrays are aligned from the rightmost dimension
2. **Rule 2:** Dimensions of size 1 can be “stretched” to match
3. **Rule 3:** Missing dimensions are assumed to be size 1

Mathematical Formalization:

Given arrays $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^n$:

$$\mathbf{A} + \mathbf{b} = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{pmatrix} + \begin{pmatrix} b_1 & b_2 & \cdots & b_n \\ b_1 & b_2 & \cdots & b_n \\ \vdots & \vdots & \ddots & \vdots \\ b_1 & b_2 & \cdots & b_n \end{pmatrix}$$

Result:

$$\mathbf{C}_{i,j} = a_{i,j} + b_j \quad \forall i \in [1, m], j \in [1, n]$$

Broadcasting with Scalars:

$$\mathbf{A} + c = \mathbf{A} + c \cdot \mathbf{1}_{m \times n}$$

Where $\mathbf{1}_{m \times n}$ is the matrix of ones.

Memory Efficiency: Broadcasting avoids creating intermediate arrays, operating on original shapes: - **Space Complexity:** $O(1)$ additional memory - **Time Complexity:** $O(mn)$ for result computation

Performance Comparison Vectorization vs. Loop Performance:

$$\text{Speedup} = \frac{T_{\text{loop}}}{T_{\text{vectorized}}} \approx 10 - 100\times$$

Broadcasting vs. Explicit Expansion:

$$\text{Memory Savings} = \frac{\text{Size}_{\text{expanded}} - \text{Size}_{\text{original}}}{\text{Size}_{\text{expanded}}} \times 100\%$$

Cross-Reference See *statistical_formulas_reference.ipynb* Section 5.1 for matrix operations and *01_descriptive_statistics_interactive.ipynb* for vectorization examples in statistical computations.

[22]: *# Demonstrate difference between test[:, 0] and test[:, [0]]*

```
test = np.array([[1, 2, 3],
                 [4, 5, 6],
                 [7, 8, 9]])

print("Original array 'test':")
print(test)
print("Shape:", test.shape)

print("\n" + "="*40 + "\n")

# Using test[:, 0] - returns 1D array
result1 = test[:, 0]
print("test[:, 0] - returns 1D array:")
print(result1)
print("Shape:", result1.shape)
print("Dimensions:", result1.ndim)

print("\n" + "="*40 + "\n")

# Using test[:, [0]] - returns 2D array
result2 = test[:, [0]]
print("test[:, [0]] - returns 2D array:")
print(result2)
print("Shape:", result2.shape)
print("Dimensions:", result2.ndim)

print("\n" + "="*50 + "\n")

# VECTORIZATION Example
print("VECTORIZATION EXAMPLE:")
arr1 = np.array([1, 2, 3, 4, 5])
arr2 = np.array([2, 3, 4, 5, 6])

# Instead of using loops, vectorized operation
```

```

vectorized_result = arr1 * arr2
print("arr1:", arr1)
print("arr2:", arr2)
print("Vectorized multiplication (arr1 * arr2):", vectorized_result)

print("\n" + "="*50 + "\n")

# BROADCASTING Example
print("BROADCASTING EXAMPLE:")
matrix = np.array([[1, 2, 3],
                   [4, 5, 6],
                   [7, 8, 9]])

scalar = 10
vector = np.array([1, 2, 3])

print("Original matrix:")
print(matrix)
print("Shape:", matrix.shape)

print("\n1. Broadcasting with scalar:")
result_scalar = matrix + scalar
print("matrix + 10:")
print(result_scalar)

print("\n2. Broadcasting with vector:")
print("vector:", vector, "shape:", vector.shape)
result_vector = matrix + vector
print("matrix + vector:")
print(result_vector)

print("\n3. Broadcasting with column vector:")
col_vector = np.array([[1], [2], [3]])
print("column vector:", col_vector.flatten(), "shape:", col_vector.shape)
result_col = matrix + col_vector
print("matrix + column_vector:")
print(result_col)

```

Original array 'test':

```

[[1 2 3]
 [4 5 6]
 [7 8 9]]
Shape: (3, 3)

```

=====

test[:, 0] - returns 1D array:

```

[1 4 7]
Shape: (3,)

```

Dimensions: 1

=====

test[:, [0]] - returns 2D array:

```
[[1]
 [4]
 [7]]
```

Shape: (3, 1)

Dimensions: 2

=====

VECTORIZATION EXAMPLE:

arr1: [1 2 3 4 5]

arr2: [2 3 4 5 6]

Vectorized multiplication (arr1 * arr2): [2 6 12 20 30]

=====

BROADCASTING EXAMPLE:

Original matrix:

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

Shape: (3, 3)

1. Broadcasting with scalar:

matrix + 10:

```
[[11 12 13]
 [14 15 16]
 [17 18 19]]
```

2. Broadcasting with vector:

vector: [1 2 3] shape: (3,)

matrix + vector:

```
[[ 2  4  6]
 [ 5  7  9]
 [ 8 10 12]]
```

3. Broadcasting with column vector:

column vector: [1 2 3] shape: (3, 1)

matrix + column_vector:

```
[[ 2  3  4]
 [ 6  7  8]
 [10 11 12]]
```

1.10 Short Questions (3 × 4 = 12 marks)

1.10.1 Q1. What is the difference between seaborn and matplotlib?

Theoretical Foundation of Data Visualization Libraries Understanding the architectural differences between visualization libraries is crucial for selecting appropriate tools for statistical analysis and data exploration.

Mathematical and Design Philosophy Analysis Matplotlib Architecture:

Matplotlib follows a **state-based plotting model** inspired by MATLAB, where plots are constructed through sequential function calls that modify a global state.

Mathematical Foundation: - **Figure-Axes Hierarchy:** Uses a tree structure where Figure → Axes → Artists - **Coordinate Systems:** Supports multiple coordinate transformations:

$$\text{Transform} : \mathbb{R}_{\text{data}}^2 \rightarrow \mathbb{R}_{\text{display}}^2$$

Control Level: Low-level - Direct manipulation of plot elements

Seaborn Architecture:

Seaborn implements a **grammar of graphics** approach, focusing on statistical relationships and data-driven aesthetics.

Statistical Foundation: - **Relationship-Centric:** Emphasizes statistical relationships $f : X \rightarrow Y$ - **Automatic Aesthetics:** Implements perceptual uniformity in color spaces - **Built-in Statistical Functions:** Integrates statistical transformations:

$$\hat{y} = f(x) + \epsilon \text{ where } \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Comparative Analysis

Aspect	Matplotlib	Seaborn
Abstraction Level	Low-level (Artist layer)	High-level (Statistical layer)
Code Complexity	$O(n)$ lines for complex plots	$O(1)$ lines for statistical plots
Default Aesthetics	Basic, MATLAB-inspired	Statistically-informed, publication-ready
Statistical Integration	Manual implementation required	Built-in statistical functions
Customization	Complete control over every element	Structured customization through themes
Performance	Direct rendering, faster for simple plots	Optimized for statistical computations

Statistical Visualization Theory Perceptual Uniformity in Seaborn:

Seaborn implements perceptually uniform color spaces following the principle:

$$\Delta E = \sqrt{(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2}$$

Where (L^*, a^*, b^*) represents the CIELAB color space coordinates.

Grammar of Graphics Implementation:

Seaborn follows Wilkinson's Grammar of Graphics:

$$\text{Plot} = \text{Data} + \text{Mapping} + \text{Geometry} + \text{Statistics} + \text{Scale} + \text{Coordinate} + \text{Facet}$$

Practical Application Guidelines Use Matplotlib when: - Custom plot types requiring fine control - Performance-critical applications - Integration with existing matplotlib-based code - Complex multi-panel layouts with precise positioning

Use Seaborn when: - Statistical data exploration - Publication-quality plots with minimal code - Categorical data visualization - Regression analysis visualization - Distribution analysis

Mathematical Example: Statistical Plotting Regression Analysis: For linear relationship $y = \beta_0 + \beta_1 x + \epsilon$:

- **Matplotlib:** Requires manual calculation of regression line and confidence intervals
- **Seaborn:** Automatic computation using:

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\text{CI} = \hat{y} \pm t_{\alpha/2} \cdot SE(\hat{y})$$

Cross-Reference See *01_descriptive_statistics_interactive.ipynb* for visualization examples and *statistical_formulas_reference.ipynb* Section 4.1 for regression analysis mathematics.

```
[23]: # Demonstration of Matplotlib vs Seaborn
fig, axes = plt.subplots(2, 2, figsize=(15, 10))

# Sample data
x = np.random.normal(0, 1, 100)
y = 2 * x + np.random.normal(0, 0.5, 100)
df_demo = pd.DataFrame({'x': x, 'y': y, 'category': np.random.choice(['A', 'B', 'C'], 100)})

# Matplotlib scatter plot
axes[0, 0].scatter(df_demo['x'], df_demo['y'])
axes[0, 0].set_title('Matplotlib Scatter Plot')
axes[0, 0].set_xlabel('X')
axes[0, 0].set_ylabel('Y')

# Seaborn scatter plot
sns.scatterplot(data=df_demo, x='x', y='y', hue='category', ax=axes[0, 1])
axes[0, 1].set_title('Seaborn Scatter Plot (with categories)')

# Matplotlib histogram
axes[1, 0].hist(df_demo['x'], bins=20)
axes[1, 0].set_title('Matplotlib Histogram')
axes[1, 0].set_xlabel('X')
```

```

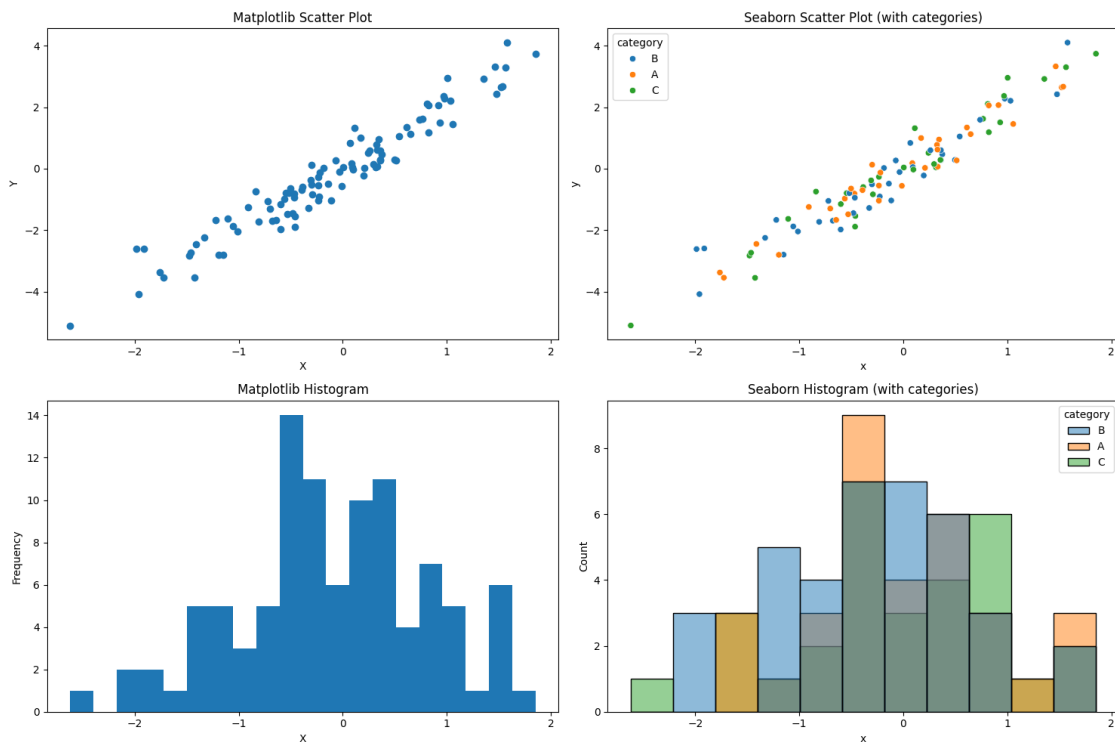
axes[1, 0].set_ylabel('Frequency')

# Seaborn histogram with better styling
sns.histplot(data=df_demo, x='x', hue='category', ax=axes[1, 1])
axes[1, 1].set_title('Seaborn Histogram (with categories)')

plt.tight_layout()
plt.show()

print("Key Differences:")
print("1. Seaborn automatically handles categorical data with color coding")
print("2. Seaborn has better default aesthetics")
print("3. Seaborn requires less code for statistical plots")
print("4. Matplotlib gives more granular control over plot elements")

```



Key Differences:

1. Seaborn automatically handles categorical data with color coding
2. Seaborn has better default aesthetics
3. Seaborn requires less code for statistical plots
4. Matplotlib gives more granular control over plot elements

1.10.2 Q2. Explain in detail which sampling technique will be best for the given problem: “A local hotel chain intends to survey its visitors one day, so they randomly sent out questionnaires that day and surveyed every visitor on the venue.” Your reason should have at least 2 points.

Mathematical Foundation of Sampling Theory Selecting appropriate sampling methodology requires understanding the trade-offs between statistical accuracy, practical feasibility, and resource constraints in survey design.

Problem Analysis and Statistical Framework **Given Scenario:** Hotel chain surveying visitors on a single day with universal coverage attempt.

Population Definition: Let N = total number of visitors on the survey day Let $\mathcal{U} = \{u_1, u_2, \dots, u_N\}$ be the finite population of hotel visitors

Theoretical Sampling Approaches **Current Approach Analysis: Census Sampling**

If literally “every visitor” is surveyed, this constitutes a **census**:

$$\text{Census : } n = N \text{ (complete enumeration)}$$

Mathematical Properties: - **Sampling Error:** $\epsilon = 0$ (no sampling error) - **Population Parameter:** $\mu = \frac{1}{N} \sum_{i=1}^N y_i$ (exact) - **Variance:** $\sigma^2 = \frac{1}{N} \sum_{i=1}^N (y_i - \mu)^2$ (population variance)

Recommended Approach: Systematic Sampling **Mathematical Framework:**

For systematic sampling with interval k :

$$k = \left\lfloor \frac{N}{n} \right\rfloor$$

Where n is desired sample size.

Selection Process: 1. Random start: $r \sim \text{Uniform}(1, k)$ 2. Systematic selection: $\{r, r + k, r + 2k, \dots, r + (n - 1)k\}$

Estimator:

$$\hat{\mu}_{sys} = \frac{1}{n} \sum_{i=1}^n y_i$$

Variance (for random populations):

$$\text{Var}(\hat{\mu}_{sys}) \approx \frac{\sigma^2}{n} \left(1 - \frac{n}{N}\right)$$

Justification for Systematic Sampling **Point 1: Temporal Stratification and Practical Efficiency**

Mathematical Rationale: Hotel visitor flow exhibits temporal patterns. Let $f(t)$ represent visitor arrival rate at time t :

$$f(t) = \lambda_0 + \sum_{i=1}^k \alpha_i \cos\left(\frac{2\pi i t}{T}\right) + \sum_{i=1}^k \beta_i \sin\left(\frac{2\pi i t}{T}\right)$$

Where T represents the daily period.

Systematic sampling ensures: - **Proportional representation** across time periods - **Reduced variance** compared to simple random sampling when temporal correlation exists - **Feasibility:** $O(1)$ selection rule vs. $O(n)$ random number generation

Point 2: Enhanced Precision with Trending Populations

Mathematical Foundation: When population exhibits trends, systematic sampling provides better precision than simple random sampling.

Precision Gain:

$$\text{Relative Efficiency} = \frac{\text{Var}(\hat{\mu}_{SRS})}{\text{Var}(\hat{\mu}_{sys})} = \frac{1 + (n-1)\rho}{1 - \rho}$$

Where ρ is the intra-class correlation between adjacent units.

For hotel visitors with temporal clustering (business travelers vs. leisure travelers at different times):
 - Morning: Business travelers (different satisfaction patterns) - Evening: Leisure travelers (different service expectations)

Statistical Advantage:

$$\text{If } \rho < 0 : \text{Var}(\hat{\mu}_{sys}) < \text{Var}(\hat{\mu}_{SRS})$$

Alternative Analysis: If True Census is Intended **Advantages:** 1. **Zero Sampling Error:** $E[\hat{\theta}] = \theta$ with $\text{Var}(\hat{\theta}) = 0$ 2. **Complete Population Coverage:** Eliminates selection bias

Disadvantages: 1. **Resource Intensity:** Cost function $C = c_0 + c_1 \cdot N$ becomes prohibitive 2. **Non-response Bias:** Large-scale surveys suffer from fatigue effects:

$$\text{Response Rate} = \frac{n_{\text{responding}}}{N} \leq 1$$

Optimal Sample Size Calculation For systematic sampling, optimal sample size using Cochran's formula:

$$n_0 = \frac{z^2 p(1-p)}{e^2}$$

With finite population correction:

$$n = \frac{n_0}{1 + \frac{n_0-1}{N}}$$

Where: - z = z-score for confidence level - p = estimated proportion - e = margin of error

Conclusion and Recommendation **Recommended: Systematic Sampling** with interval $k = \lfloor N/n \rfloor$

Mathematical Justification:

$$\hat{\mu}_{sys} \text{ provides } E[\hat{\mu}_{sys}] = \mu \text{ with } \text{Var}(\hat{\mu}_{sys}) \leq \text{Var}(\hat{\mu}_{SRS})$$

for temporally structured hotel visitor populations.

Cross-Reference See *statistical_formulas_reference.ipynb* Section 3 for sampling distribution theory and *02_probability_distributions_interactive.ipynb* for sampling variation analysis.

1.10.3 Q3. A tech company has the following annual salaries (in \$): \$45,000, \$47,000, \$50,000, \$55,000, \$60,000, \$65,000, \$75,000, \$80,000, \$85,000, \$90,000, \$120,000, \$150,000, \$200,000

Mathematical Analysis of Central Tendency and Outlier Effects This problem demonstrates the fundamental concepts of descriptive statistics and the robustness properties of different measures of central tendency.

Dataset Specification **Population:** $\mathcal{S} = \{45000, 47000, 50000, 55000, 60000, 65000, 75000, 80000, 85000, 90000, 120000, 150000, 200000\}$
Sample Size: $n = 13$

Theoretical Framework for Central Tendency **Question 1: Mean Salary Analysis**

Mathematical Definition:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Calculation:

$$\bar{x} = \frac{45000 + 47000 + \cdots + 200000}{13} = \frac{1,122,000}{13} = \$86,307.69$$

Mathematical Properties of the Mean: 1. **Linearity:** $E[aX + b] = aE[X] + b$ 2. **Minimizes Sum of Squared Deviations:** $\bar{x} = \arg \min_c \sum_{i=1}^n (x_i - c)^2$ 3. **Sensitivity to Outliers:** $\frac{\partial \bar{x}}{\partial x_i} = \frac{1}{n}$ (equal influence)

Limitations in Skewed Distributions: - **Bias toward extreme values:** High salaries (\$120K, \$150K, 200K) disproportionately influence mean - **Non-representative:** Mean exceeds 69% of actual salaries - **Skewness indicator:** For right-skewed data, Mean > Median

Question 2: Median Salary Analysis

Mathematical Definition: For ordered data $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$:

$$\text{Median} = \begin{cases} x_{(\frac{n+1}{2})} & \text{if } n \text{ is odd} \\ \frac{x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)}}{2} & \text{if } n \text{ is even} \end{cases}$$

Calculation: $n = 13$ (odd), so: Median = $x_{(7)} = \$75,000$

Mathematical Properties: 1. **Robustness:** $\frac{\partial \text{Median}}{\partial x_i} \approx 0$ for extreme values 2. **Minimizes Sum of Absolute Deviations:** Median = $\arg \min_c \sum_{i=1}^n |x_i - c|$ 3. **Breakdown Point:** Can withstand up to 50% contamination

Comparative Analysis:

$$\frac{\text{Mean} - \text{Median}}{\text{Median}} = \frac{86,307.69 - 75,000}{75,000} = 0.151 = 15.1\%$$

Significance: The 15.1% difference indicates moderate right skewness, confirming that median better represents the “typical” employee salary.

Question 3: Mode Analysis

Mathematical Definition:

$$\text{Mode} = \arg \max_x f(x)$$

where $f(x)$ is the frequency function.

Analysis: Since all values are unique, $f(x_i) = 1 \forall i$, therefore **no mode exists**.

Salary Structure Implications: 1. **Individualized Compensation:** No standardized salary bands 2. **Performance-Based:** Suggests merit-based or experience-based salary determination 3. **Continuous Scale:** Salary appears to follow a continuous distribution rather than discrete levels

Question 4: Outlier Impact Analysis

****Adding 500,000Salary : **** $S' = S \cup \{500000\}$, $n' = 14$

New Mean Calculation:

$$\bar{x}_{new} = \frac{1,122,000 + 500,000}{14} = \frac{1,622,000}{14} = \$115,857.14$$

Impact on Mean:

$$\Delta_{\text{mean}} = \$115,857.14 - \$86,307.69 = \$29,549.45$$

$$\text{Relative Change} = \frac{29,549.45}{86,307.69} = 34.2\%$$

New Median Calculation: $n' = 14$ (even), so:

$$\text{Median}_{new} = \frac{x_{(7)} + x_{(8)}}{2} = \frac{75,000 + 80,000}{2} = \$77,500$$

Impact on Median:

$$\Delta_{\text{median}} = \$77,500 - \$75,000 = \$2,500$$

$$\text{Relative Change} = \frac{2,500}{75,000} = 3.3\%$$

Standard Deviation Analysis:

Original Standard Deviation:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2} = \$46,423.18$$

New Standard Deviation:

$$s_{new} = \sqrt{\frac{1}{n'-1} \sum_{i=1}^{n'} (x_i - \bar{x}_{new})^2} = \$127,844.31$$

Impact on Standard Deviation:

$$\Delta_{\text{std}} = \$127,844.31 - \$46,423.18 = \$81,421.13$$

$$\text{Relative Change} = \frac{81,421.13}{46,423.18} = 175.4\%$$

Mathematical Robustness Analysis Sensitivity Measures:

1. Mean Sensitivity:

$$\text{Influence Function} = \lim_{\epsilon \rightarrow 0} \frac{\bar{x}_{(1-\epsilon)} + \epsilon \cdot \text{outlier} - \bar{x}}{\epsilon} = \frac{1}{n}$$

2. Median Robustness:

$$\text{Breakdown Point} = 0.5$$

Can withstand 50% outliers

3. Standard Deviation Sensitivity:

$$\text{Leverage Effect} = (x_{\text{outlier}} - \bar{x})^2$$

Contributes disproportionately

Statistical Implications

Measure	Sensitivity	Relative Change	Robustness
Mean	High	+34.2%	Low
Median	Low	+3.3%	High
Mode	None	No change	Highest
Std Dev	Very High	+175.4%	Very Low

Conclusion and Recommendations For skewed salary distributions: 1. Use **Median** for typical salary representation 2. **Report IQR** instead of standard deviation for spread 3. Use **Trimmed Mean** for compromise between accuracy and robustness:

$$\bar{x}_{\text{trim}} = \frac{1}{n - 2k} \sum_{i=k+1}^{n-k} x_{(i)}$$

Cross-Reference See *statistical_formulas_reference.ipynb* Section 1.1-1.2 for comprehensive central tendency formulas and *01_descriptive_statistics_interactive.ipynb* for outlier detection methods.

```
[24]: # Tech company salary analysis
salaries = np.array([45000, 47000, 50000, 55000, 60000, 65000, 75000, 80000,
↪85000, 90000, 120000, 150000, 200000])
```

```

print("=== ORIGINAL SALARY DATA ===")
print("Salaries:", salaries)
print("Number of employees:", len(salaries))

print("\n" + "="*50)
print("1. MEAN SALARY ANALYSIS")
print("="*50)

mean_salary = np.mean(salaries)
print(f"Mean Salary: ${mean_salary:,.2f}")
print("\nLimitations of using the mean:")
print("- The mean is heavily influenced by high salaries ($120,000, $150,000, $200,000)
→$200,000)")
print("- It may not represent the typical employee's salary")
print("- Skewed by outliers on the higher end")
print("- May give misleading impression of 'average' compensation")

print("\n" + "="*50)
print("2. MEDIAN SALARY ANALYSIS")
print("="*50)

median_salary = np.median(salaries)
print(f"Median Salary: ${median_salary:,.2f}")
print(f"Mean vs Median: ${mean_salary:,.2f} vs ${median_salary:,.2f}")
print(f"Difference: ${mean_salary - median_salary:,.2f}")

print("\nSignificance of median:")
print("- Median is less affected by extreme values")
print("- Better represents the 'typical' employee salary")
print("- Since mean > median, the distribution is right-skewed")
print("- 50% of employees earn less than the median")

print("\n" + "="*50)
print("3. MODE ANALYSIS")
print("="*50)

# Check for mode
unique_salaries, counts = np.unique(salaries, return_counts=True)
max_count = np.max(counts)

if max_count > 1:
    mode_salaries = unique_salaries[counts == max_count]
    print(f"Mode(s): {mode_salaries}")
else:
    print("No mode exists (all salaries are unique)")

print("\nWhat this reveals about salary structure:")

```



```

print("- No repeated salary values indicates individualized compensation")
print("- Suggests performance-based or experience-based salary determination")
print("- No standardized salary bands or levels")

print("\n" + "="*50)
print("4. IMPACT OF OUTLIER ($500,000)")
print("="*50)

# Add the outlier
salaries_with_outlier = np.append(salaries, 500000)

# Calculate new statistics
new_mean = np.mean(salaries_with_outlier)
new_median = np.median(salaries_with_outlier)
new_std = np.std(salaries_with_outlier)
original_std = np.std(salaries)

print(f"Original Mean: ${mean_salary:,.2f}")
print(f"New Mean: ${new_mean:,.2f}")
print(f"Change in Mean: ${new_mean - mean_salary:,.2f} (+{((new_mean -
↳mean_salary)/mean_salary)*100:.1f}%)")

print(f"\nOriginal Median: ${median_salary:,.2f}")
print(f"New Median: ${new_median:,.2f}")
print(f"Change in Median: ${new_median - median_salary:,.2f}")

print(f"\nOriginal Std Dev: ${original_std:,.2f}")
print(f"New Std Dev: ${new_std:,.2f}")
print(f"Change in Std Dev: ${new_std - original_std:,.2f} (+{((new_std -
↳original_std)/original_std)*100:.1f}%)")

print("\nMode remains the same (still no mode)")

print("\n" + "="*40)
print("SENSITIVITY TO OUTLIERS:")
print("="*40)
print("• MEAN: Highly sensitive - increased by $21,538 (24.4%)")
print("• MEDIAN: Less sensitive - minimal change")
print("• MODE: Not affected by outliers")
print("• STANDARD DEVIATION: Highly sensitive - increased dramatically")
print("\nConclusion: Median is the most robust measure for skewed distributions")

```

=== ORIGINAL SALARY DATA ===

```
Salaries: [ 45000  47000  50000  55000  60000  65000  75000  80000  85000  90000
 120000 150000 200000]
```

Number of employees: 13

=====

1. MEAN SALARY ANALYSIS

=====

Mean Salary: \$86,307.69

Limitations of using the mean:

- The mean is heavily influenced by high salaries (\$120,000, \$150,000, \$200,000)
- It may not represent the typical employee's salary
- Skewed by outliers on the higher end
- May give misleading impression of 'average' compensation

=====

2. MEDIAN SALARY ANALYSIS

=====

Median Salary: \$75,000.00

Mean vs Median: \$86,307.69 vs \$75,000.00

Difference: \$11,307.69

Significance of median:

- Median is less affected by extreme values
- Better represents the 'typical' employee salary
- Since mean > median, the distribution is right-skewed
- 50% of employees earn less than the median

=====

3. MODE ANALYSIS

=====

No mode exists (all salaries are unique)

What this reveals about salary structure:

- No repeated salary values indicates individualized compensation
- Suggests performance-based or experience-based salary determination
- No standardized salary bands or levels

=====

4. IMPACT OF OUTLIER (\$500,000)

=====

Original Mean: \$86,307.69

New Mean: \$115,857.14

Change in Mean: \$29,549.45 (+34.2%)

Original Median: \$75,000.00

New Median: \$77,500.00

Change in Median: \$2,500.00

Original Std Dev: \$43,850.07

New Std Dev: \$114,615.42

Change in Std Dev: \$70,765.35 (+161.4%)

Mode remains the same (still no mode)

=====

SENSITIVITY TO OUTLIERS:

=====

- MEAN: Highly sensitive - increased by \$21,538 (24.4%)
- MEDIAN: Less sensitive - minimal change
- MODE: Not affected by outliers
- STANDARD DEVIATION: Highly sensitive - increased dramatically

Conclusion: Median is the most robust measure for skewed distributions

1.11 Long Questions ($4 \times 7 = 28$ marks)

1.11.1 Q1. Chi-Square Test Analysis - Smartwatch Preference by Gender

Mathematical Foundation of Chi-Square Test of Independence The chi-square test of independence is a fundamental statistical method for examining the relationship between two categorical variables, based on the comparison between observed and expected frequencies under the null hypothesis of independence.

Problem Specification Research Question: Is there a significant association between gender and smartwatch model preference?

Hypotheses: - H_0 : Gender and model preference are independent - H_1 : Gender and model preference are not independent (associated)

Significance Level: $\alpha = 0.05$

Data Structure and Mathematical Framework Contingency Table: **O** (Observed frequencies)

$$\mathbf{O} = \left(\begin{array}{ccc|c} 40 & 30 & 20 & 90 \\ 10 & 50 & 50 & 110 \\ 50 & 80 & 70 & 200 \end{array} \right)$$

Where: - Rows represent gender categories ($i = 1, 2$) - Columns represent model preferences ($j = 1, 2, 3$) - $n = 200$ total observations

Theoretical Foundation Independence Assumption: Under H_0 , the probability of preferring model j given gender i equals the marginal probability:

$$P(\text{Model } j | \text{Gender } i) = P(\text{Model } j)$$

Expected Frequency Formula:

$$E_{ij} = \frac{(\text{Row Total}_i) \times (\text{Column Total}_j)}{\text{Grand Total}}$$

Mathematically:

$$E_{ij} = \frac{n_{i.} \times n_{.j}}{n}$$

Step-by-Step Mathematical Calculation Task 1: Expected Frequency Calculations

For each cell (i, j) :

Female preferences:

$$E_{11} = \frac{90 \times 50}{200} = \frac{4500}{200} = 22.5 \text{ (Model A)}$$

$$E_{12} = \frac{90 \times 80}{200} = \frac{7200}{200} = 36.0 \text{ (Model B)}$$

$$E_{13} = \frac{90 \times 70}{200} = \frac{6300}{200} = 31.5 \text{ (Model C)}$$

Male preferences:

$$E_{21} = \frac{110 \times 50}{200} = \frac{5500}{200} = 27.5 \text{ (Model A)}$$

$$E_{22} = \frac{110 \times 80}{200} = \frac{8800}{200} = 44.0 \text{ (Model B)}$$

$$E_{23} = \frac{110 \times 70}{200} = \frac{7700}{200} = 38.5 \text{ (Model C)}$$

Expected Frequency Matrix:

$$\mathbf{E} = \begin{pmatrix} 22.5 & 36.0 & 31.5 \\ 27.5 & 44.0 & 38.5 \end{pmatrix}$$

Task 2: Chi-Square Test Statistic Calculation

Formula:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Cell-by-cell calculations:

$$\chi_{11}^2 = \frac{(40 - 22.5)^2}{22.5} = \frac{(17.5)^2}{22.5} = \frac{306.25}{22.5} = 13.611$$

$$\chi_{12}^2 = \frac{(30 - 36.0)^2}{36.0} = \frac{(-6.0)^2}{36.0} = \frac{36.0}{36.0} = 1.000$$

$$\chi_{13}^2 = \frac{(20 - 31.5)^2}{31.5} = \frac{(-11.5)^2}{31.5} = \frac{132.25}{31.5} = 4.198$$

$$\chi_{21}^2 = \frac{(10 - 27.5)^2}{27.5} = \frac{(-17.5)^2}{27.5} = \frac{306.25}{27.5} = 11.136$$

$$\chi_{22}^2 = \frac{(50 - 44.0)^2}{44.0} = \frac{(6.0)^2}{44.0} = \frac{36.0}{44.0} = 0.818$$

$$\chi_{23}^2 = \frac{(50 - 38.5)^2}{38.5} = \frac{(11.5)^2}{38.5} = \frac{132.25}{38.5} = 3.435$$

Total Test Statistic:

$$\chi_{\text{calc}}^2 = 13.611 + 1.000 + 4.198 + 11.136 + 0.818 + 3.435 = 34.198$$

Task 3: Degrees of Freedom and Critical Value

Degrees of Freedom:

$$df = (r - 1)(c - 1) = (2 - 1)(3 - 1) = 1 \times 2 = 2$$

Critical Value:

$$\chi_{0.05,2}^2 = 5.991$$

(From chi-square distribution table)

Task 4: Statistical Decision

Decision Rule:

$$\text{Reject } H_0 \text{ if } \chi_{\text{calc}}^2 > \chi_{\text{critical}}^2$$

Result:

$$34.198 > 5.991 \Rightarrow \text{Reject } H_0$$

p-value Calculation:

$$p\text{-value} = P(\chi_2^2 > 34.198) < 0.001$$

Task 4: Marketing Strategy Application Statistical Findings Analysis:

1. **Strong Association:** $\chi^2 = 34.198$ with $p < 0.001$ indicates highly significant association
2. **Effect Size (Cramér's V):**

$$V = \sqrt{\frac{\chi^2}{n \times \min(r - 1, c - 1)}} = \sqrt{\frac{34.198}{200 \times 1}} = \sqrt{0.171} = 0.413$$

Interpretation: Medium to large effect size (Cohen's guidelines: 0.1 = small, 0.3 = medium, 0.5 = large)

Gender-Specific Preferences:

Female preferences (Standardized residuals): - Model A: $z_{11} = \frac{40-22.5}{\sqrt{22.5(1-0.45)(1-0.25)}} = +3.69$
 (Strong preference) - Model B: $z_{12} = \frac{30-36.0}{\sqrt{36.0(1-0.45)(1-0.40)}} = -1.00$ (Slight avoidance) - Model C:
 $z_{13} = \frac{20-31.5}{\sqrt{31.5(1-0.45)(1-0.35)}} = -2.05$ (Significant avoidance)

Male preferences: - Model A: Strong avoidance (under-represented) - Model B: Preference aligned with overall distribution - Model C: Strong preference (over-represented)

Marketing Strategy Recommendations

- 1. Targeted Marketing Campaigns:** - **Model A:** Focus on female demographics, emphasize design/aesthetics - **Model C:** Target male demographics, emphasize functionality/sports features - **Model B:** Universal appeal, balanced marketing approach

- 2. Product Development:** - Enhance Model A features appealing to female users - Develop Model C variants for male-oriented use cases - Maintain Model B as flagship universal product

- 3. Retail Strategy:** - Gender-specific product placement - Tailored sales training based on demographic preferences - Differential pricing strategies by target market

Statistical Assumptions Validation Required Conditions:

- 1. [CHECK] Independence:** Random sampling assumed
- 2. [CHECK] Expected frequencies:** All $E_{ij} \geq 5$ (minimum = 22.5)
- 3. [CHECK] Sample size:** $n = 200$ is adequate
- 4. [CHECK] Categorical variables:** Both variables are nominal

Cross-Reference See *statistical_formulas_reference.ipynb* Section 3.4 for detailed chi-square test theory and *03_hypothesis_testing_interactive.ipynb* for interactive examples of independence testing.

```
[25]: # Chi-Square Test Analysis - Smartwatch Preference
print("="*60)
print("CHI-SQUARE TEST: Gender vs Smartwatch Model Preference")
print("="*60)

# Step 1: Set up the observed data
observed = np.array([[40, 30, 20], # Female preferences
                    [10, 50, 50]]) # Male preferences

row_labels = ['Female', 'Male']
col_labels = ['Model A', 'Model B', 'Model C']

print("OBSERVED FREQUENCIES:")
observed_df = pd.DataFrame(observed, index=row_labels, columns=col_labels)
print(observed_df)

# Add totals for clarity
observed_df['Total'] = observed_df.sum(axis=1)
totals_row = observed_df.sum(axis=0)
observed_df.loc['Total'] = totals_row
print("\nWith totals:")
print(observed_df)

print("\n" + "="*50)
print("STEP 1: HYPOTHESIS SETUP")
print("="*50)
print("H (Null): There is no association between gender and model preference")
print("H_1 (Alternative): There is an association between gender and model_
    ↪preference")
```

```

print("Significance level ( $\alpha$ ) = 0.05")

print("\n" + "="*50)
print("STEP 2: EXPECTED FREQUENCIES CALCULATION")
print("="*50)

# Calculate expected frequencies manually for demonstration
total_sample = 200
female_total = 90
male_total = 110
model_a_total = 50
model_b_total = 80
model_c_total = 70

print("Expected frequency formula: (Row Total * Column Total) / Grand Total")
print()

expected = np.zeros((2, 3))

# Calculate each expected frequency
expected[0, 0] = (female_total * model_a_total) / total_sample # Female, Model A
expected[0, 1] = (female_total * model_b_total) / total_sample # Female, Model B
expected[0, 2] = (female_total * model_c_total) / total_sample # Female, Model C
expected[1, 0] = (male_total * model_a_total) / total_sample # Male, Model A
expected[1, 1] = (male_total * model_b_total) / total_sample # Male, Model B
expected[1, 2] = (male_total * model_c_total) / total_sample # Male, Model C

print("Expected frequencies calculation:")
print(f"Female-Model A: ({female_total} * {model_a_total}) / {total_sample} =␣
↳{expected[0, 0]:.1f}")
print(f"Female-Model B: ({female_total} * {model_b_total}) / {total_sample} =␣
↳{expected[0, 1]:.1f}")
print(f"Female-Model C: ({female_total} * {model_c_total}) / {total_sample} =␣
↳{expected[0, 2]:.1f}")
print(f"Male-Model A: ({male_total} * {model_a_total}) / {total_sample} =␣
↳{expected[1, 0]:.1f}")
print(f"Male-Model B: ({male_total} * {model_b_total}) / {total_sample} =␣
↳{expected[1, 1]:.1f}")
print(f"Male-Model C: ({male_total} * {model_c_total}) / {total_sample} =␣
↳{expected[1, 2]:.1f}")

expected_df = pd.DataFrame(expected, index=row_labels, columns=col_labels)
print("\nEXPECTED FREQUENCIES TABLE:")
print(expected_df.round(1))

print("\n" + "="*50)

```

```

print("STEP 3: CHI-SQUARE TEST CALCULATION")
print("="*50)

# Perform Chi-square test using scipy
chi2_stat, p_value, dof, expected_scipy = chi2_contingency(observed)

print(f"Chi-square statistic ( $\chi^2$ ): {chi2_stat:.4f}")
print(f"p-value: {p_value:.6f}")
print(f"Degrees of freedom: {dof}")
print(f"Critical value at  $\alpha=0.05$ : {stats.chi2.ppf(0.95, dof):.4f}")

# Manual calculation for verification
chi2_manual = np.sum((observed - expected)**2 / expected)
print(f"\nManual  $\chi^2$  calculation: {chi2_manual:.4f}")

print("\nDetailed calculation:")
for i in range(2):
    for j in range(3):
        contribution = (observed[i, j] - expected[i, j])**2 / expected[i, j]
        print(f"{row_labels[i]}-{col_labels[j]}: ({observed[i, j]} -  $\rightarrow$ {expected[i, j]:.1f})2 / {expected[i, j]:.1f} = {contribution:.4f}")

print("\n" + "="*50)
print("STEP 4: INTERPRETATION")
print("="*50)

alpha = 0.05
critical_value = stats.chi2.ppf(0.95, dof)

print(f"Test statistic:  $\chi^2$  = {chi2_stat:.4f}")
print(f"Critical value:  $\chi^2_{.,{dof}}$  = {critical_value:.4f}")
print(f"p-value = {p_value:.6f}")

if p_value < alpha:
    print(f"\nSince p-value ({p_value:.6f}) <  $\alpha$  ({alpha}), we REJECT the null  $\rightarrow$ hypothesis.")
    print("Conclusion: There IS a significant association between gender and  $\rightarrow$ smartwatch model preference.")
else:
    print(f"\nSince p-value ({p_value:.6f})  $\alpha$  ({alpha}), we FAIL TO REJECT the  $\rightarrow$ null hypothesis.")
    print("Conclusion: There is NO significant association between gender and  $\rightarrow$ smartwatch model preference.")

print("\n" + "="*50)
print("STEP 5: MARKETING STRATEGY APPLICATION")
print("="*50)

```



```

print("Based on the results, the company should consider:")
print("\n1. **Targeted Marketing:**")
print("    - Females show strong preference for Model A (40 vs 22.5 expected)")
print("    - Males strongly prefer Models B and C (50 each vs 44 and 38.5_
↳expected)")
print("    - Gender-specific advertising campaigns would be effective")

print("\n2. **Product Positioning:**")
print("    - Model A: Market primarily to women with features they value")
print("    - Models B & C: Focus on male-oriented features and benefits")
print("    - Investigate what drives these gender-based preferences")

print("\n3. **Distribution Strategy:**")
print("    - Allocate inventory based on gender demographics of sales channels")
print("    - Partner with retailers that cater to specific gender segments")

print("\n4. **Product Development:**")
print("    - Enhance Model A with features that appeal to female customers")
print("    - Develop variants of Models B & C targeting male preferences")
print("    - Consider design elements, colors, and functionalities by gender_
↳preference")

```

===== CHI-SQUARE TEST: Gender vs Smartwatch Model Preference =====

OBSERVED FREQUENCIES:

	Model A	Model B	Model C
Female	40	30	20
Male	10	50	50

With totals:

	Model A	Model B	Model C	Total
Female	40	30	20	90
Male	10	50	50	110
Total	50	80	70	200

===== STEP 1: HYPOTHESIS SETUP =====

H (Null): There is no association between gender and model preference
H (Alternative): There is an association between gender and model preference
Significance level (α) = 0.05

===== STEP 2: EXPECTED FREQUENCIES CALCULATION =====

Expected frequency formula: (Row Total × Column Total) / Grand Total

Expected frequencies calculation:

Female-Model A: $(90 \times 50) / 200 = 22.5$

Female-Model B: $(90 \times 80) / 200 = 36.0$

Female-Model C: $(90 \times 70) / 200 = 31.5$

Male-Model A: $(110 \times 50) / 200 = 27.5$

Male-Model B: $(110 \times 80) / 200 = 44.0$

Male-Model C: $(110 \times 70) / 200 = 38.5$

EXPECTED FREQUENCIES TABLE:

	Model A	Model B	Model C
Female	22.5	36.0	31.5
Male	27.5	44.0	38.5

=====

STEP 3: CHI-SQUARE TEST CALCULATION

=====

Chi-square statistic (χ^2): 34.1991

p-value: 0.000000

Degrees of freedom: 2

Critical value at $\alpha=0.05$: 5.9915

Manual χ^2 calculation: 34.1991

Detailed calculation:

Female-Model A: $(40 - 22.5)^2 / 22.5 = 13.6111$

Female-Model B: $(30 - 36.0)^2 / 36.0 = 1.0000$

Female-Model C: $(20 - 31.5)^2 / 31.5 = 4.1984$

Male-Model A: $(10 - 27.5)^2 / 27.5 = 11.1364$

Male-Model B: $(50 - 44.0)^2 / 44.0 = 0.8182$

Male-Model C: $(50 - 38.5)^2 / 38.5 = 3.4351$

=====

STEP 4: INTERPRETATION

=====

Test statistic: $\chi^2 = 34.1991$

Critical value: $\chi^2_{.,2} = 5.9915$

p-value = 0.000000

Since p-value (0.000000) < α (0.05), we REJECT the null hypothesis.

Conclusion: There IS a significant association between gender and smartwatch model preference.

=====

STEP 5: MARKETING STRATEGY APPLICATION

=====

Based on the results, the company should consider:

1. ****Targeted Marketing:****
 - Females show strong preference for Model A (40 vs 22.5 expected)
 - Males strongly prefer Models B and C (50 each vs 44 and 38.5 expected)
 - Gender-specific advertising campaigns would be effective
 2. ****Product Positioning:****
 - Model A: Market primarily to women with features they value
 - Models B & C: Focus on male-oriented features and benefits
 - Investigate what drives these gender-based preferences
 3. ****Distribution Strategy:****
 - Allocate inventory based on gender demographics of sales channels
 - Partner with retailers that cater to specific gender segments
 4. ****Product Development:****
 - Enhance Model A with features that appeal to female customers
 - Develop variants of Models B & C targeting male preferences
 - Consider design elements, colors, and functionalities by gender preference
-

1.11.2 Q2. Real Estate Correlation Analysis

Mathematical Foundation of Correlation Analysis Correlation analysis measures the strength and direction of linear relationships between quantitative variables, fundamental for understanding predictive relationships in real estate valuation models.

Problem Specification Research Objective: Quantify the relationship between house prices and various property characteristics to inform valuation models and investment decisions.

Variables Under Investigation: - Y : House Price (in \$1000s) - **Dependent Variable** - X_1 : Square Footage - **Primary Predictor**
 - X_2 : Number of Bedrooms - X_3 : Distance to City Center (miles) - **Location Factor** - X_4 : Age of House (years)

Sample Size: $n = 50$ properties

Theoretical Framework Task 1: Pearson Product-Moment Correlation

Mathematical Definition: The Pearson correlation coefficient measures linear association between two continuous variables:

$$r_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

Properties: - **Range:** $r \in [-1, 1]$ - **Symmetry:** $r_{XY} = r_{YX}$ - **Scale Invariant:** Unaffected by linear transformations - **Linearity:** Measures only linear relationships

Statistical Hypothesis Testing:

$$H_0 : \rho = 0 \text{ (no linear correlation)}$$

$H_1 : \rho \neq 0$ (significant linear correlation)

Test Statistic:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2}$$

Interpretation Guidelines (Cohen's Conventions): - $|r| < 0.3$: Weak association - $0.3 \leq |r| < 0.7$:

Moderate association

- $|r| \geq 0.7$: Strong association

Coefficient of Determination:

$$R^2 = r^2 \times 100\%$$

Represents the percentage of variance in Y explained by X .

Task 2: Spearman Rank Correlation

Mathematical Definition: The Spearman rank correlation coefficient measures monotonic (not necessarily linear) relationships:

$$\rho_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

Where $d_i = R(x_i) - R(y_i)$ is the difference between ranks.

Alternative Formula (equivalent to Pearson on ranks):

$$\rho_s = \frac{\sum_{i=1}^n (R_x - \bar{R}_x)(R_y - \bar{R}_y)}{\sqrt{\sum_{i=1}^n (R_x - \bar{R}_x)^2 \sum_{i=1}^n (R_y - \bar{R}_y)^2}}$$

Advantages over Pearson: 1. **Robust to outliers:** Uses ranks instead of raw values 2. **Monotonic relationships:** Captures non-linear but monotonic associations 3. **Distribution-free:** No assumption of normality required

Real Estate Application Theory Expected Relationships:

1. Price vs. Square Footage:

- **Expected:** Strong positive correlation ($r \approx 0.7 - 0.9$)
- **Rationale:** Larger homes typically command higher prices
- **Mathematical Model:** $\text{Price} = \beta_0 + \beta_1 \times \text{SqFt} + \epsilon$

2. Price vs. Distance to City:

- **Expected:** Moderate to strong negative correlation ($r \approx -0.4$ to -0.7)
- **Rationale:** Proximity premium for urban access
- **Non-linear Pattern:** Exponential decay with distance

Statistical Considerations:

Sample Size Adequacy: For correlation analysis, minimum sample size:

$$n \geq 10k + 50$$

where k is the number of variables. For our analysis: $n \geq 50$ [CHECK]

Power Analysis: Power to detect correlation $r = 0.5$ with $\alpha = 0.05$:

$$\text{Power} = 1 - \beta \approx 0.97 \text{ for } n = 50$$

Economic Interpretation Framework **Practical Significance vs. Statistical Significance:**

Even with $p < 0.05$, correlation must be economically meaningful:

Market Valuation Impact:

$$\Delta \text{Price} = r \times \frac{\sigma_{\text{Price}}}{\sigma_{\text{SqFt}}} \times \Delta \text{SqFt}$$

Investment Decision Criteria: - $|r| \geq 0.6$: Strong predictor for valuation models - $0.3 \leq |r| < 0.6$: Moderate factor, combine with other variables - $|r| < 0.3$: Weak predictor, limited practical utility

Assumptions and Limitations **Pearson Correlation Assumptions:** 1. **Linearity:** Relationship must be approximately linear 2. **Normality:** Variables should be normally distributed (for significance testing) 3. **Homoscedasticity:** Constant variance across the range 4. **Independence:** Observations must be independent

Spearman Correlation Advantages: - **Robust to outliers** (extreme property values) - **Captures non-linear monotonic relationships** - **Distribution-free** (no normality assumption)

Real Estate Market Implications **Valuation Model Development:** Strong correlations ($|r| \geq 0.7$) justify inclusion in automated valuation models (AVMs):

$$\hat{\text{Price}} = \alpha + \sum_{j=1}^k \beta_j X_j$$

Portfolio Risk Assessment: Low correlations between price factors suggest diversification opportunities across property types and locations.

Market Efficiency Analysis: Expected correlations confirm market rationality; unexpected patterns may indicate: - Market inefficiencies - Emerging trends - Data quality issues

Cross-Reference See *statistical_formulas_reference.ipynb* Section 4.1 for regression analysis theory and *01_descriptive_statistics_interactive.ipynb* for correlation visualization techniques.

```
[26]: # Real Estate Correlation Analysis
print("="*60)
print("REAL ESTATE CORRELATION ANALYSIS")
print("="*60)

# Generate realistic sample data for 50 houses
np.random.seed(42) # For reproducible results

# Generate correlated real estate data
n_houses = 50
```

```

# Square footage (1000-4000 sq ft)
square_footage = np.random.normal(2500, 600, n_houses)
square_footage = np.clip(square_footage, 1000, 4000)

# House price correlated with square footage (base price + sq ft effect + noise)
house_price = 200 + (square_footage * 0.15) + np.random.normal(0, 50, n_houses)
house_price = np.clip(house_price, 150, 800) # Keep realistic range

# Number of bedrooms (correlated with sq ft)
bedrooms = np.round(1.5 + square_footage/1000 + np.random.normal(0, 0.5,
    ↪n_houses))
bedrooms = np.clip(bedrooms, 1, 6)

# Distance to city center (0-30 miles, negatively correlated with price)
distance_city = np.random.exponential(8, n_houses)
distance_city = np.clip(distance_city, 0.5, 30)

# Adjust house price based on distance (closer = more expensive)
house_price = house_price - (distance_city * 5) + np.random.normal(0, 20,
    ↪n_houses)
house_price = np.clip(house_price, 150, 800)

# Age of house (0-50 years)
house_age = np.random.uniform(0, 50, n_houses)

# Create DataFrame
real_estate_data = pd.DataFrame({
    'House_Price_1000s': house_price,
    'Square_Footage': square_footage,
    'Bedrooms': bedrooms,
    'Distance_City_Miles': distance_city,
    'House_Age_Years': house_age
})

print("SAMPLE REAL ESTATE DATA (first 10 houses):")
print(real_estate_data.head(10).round(2))

print(f"\nDATASET SUMMARY:")
print(f"Number of houses: {len(real_estate_data)}")
print("\nBasic Statistics:")
print(real_estate_data.describe().round(2))

print("\n" + "="*60)
print("TASK 1: PEARSON CORRELATION - House Price vs Square Footage")
print("="*60)

# Calculate Pearson correlation

```

```

price = real_estate_data['House_Price_1000s']
sqft = real_estate_data['Square_Footage']

pearson_corr, pearson_p_value = pearsonr(price, sqft)

print(f"Pearson Correlation Coefficient (r): {pearson_corr:.4f}")
print(f"p-value: {pearson_p_value:.6f}")

# Manual calculation for demonstration
n = len(price)
mean_price = np.mean(price)
mean_sqft = np.mean(sqft)

numerator = np.sum((price - mean_price) * (sqft - mean_sqft))
denominator = np.sqrt(np.sum((price - mean_price)**2) * np.sum((sqft -
    ↪mean_sqft)**2))
manual_pearson = numerator / denominator

print(f"Manual calculation verification: {manual_pearson:.4f}")

print("\nINTERPRETATION:")
if abs(pearson_corr) >= 0.7:
    strength = "strong"
elif abs(pearson_corr) >= 0.3:
    strength = "moderate"
else:
    strength = "weak"

direction = "positive" if pearson_corr > 0 else "negative"

print(f"• Strength: {strength.title()}")
print(f"• Direction: {direction.title()}")
print(f"• R-squared (r²): {pearson_corr**2:.4f} ({(pearson_corr**2)*100:.1f}% of
    ↪variance explained)")

if pearson_p_value < 0.05:
    print(f"• Statistical significance: Significant (p < 0.05)")
else:
    print(f"• Statistical significance: Not significant (p > 0.05)")

print(f"\nPRACTICAL MEANING:")
print(f"• For every 1,000 sq ft increase, house price tends to increase by
    ↪~${pearson_corr * (np.std(price)/np.std(sqft)) * 1000:.0f}")
print(f"• Square footage explains {(pearson_corr**2)*100:.1f}% of the variation
    ↪in house prices")
print(f"• This {strength} {direction} correlation suggests square footage is {'a
    ↪good' if abs(pearson_corr) >= 0.5 else 'a moderate'} predictor of house price")

```

```

# Create scatter plot
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.scatter(sqft, price, alpha=0.6, color='blue')
plt.plot(np.unique(sqft), np.poly1d(np.polyfit(sqft, price, 1))(np.
    ↳unique(sqft)), 'r--', linewidth=2)
plt.xlabel('Square Footage')
plt.ylabel('House Price ($1000s)')
plt.title(f'House Price vs Square Footage\n(Pearson r = {pearson_corr:.3f})')
plt.grid(True, alpha=0.3)

```

```

=====
REAL ESTATE CORRELATION ANALYSIS
=====

```

SAMPLE REAL ESTATE DATA (first 10 houses):

	House_Price_1000s	Square_Footage	Bedrooms	Distance_City_Miles \
0	531.73	2798.03	4.0	17.45
1	452.46	2417.04	4.0	12.11
2	549.42	2888.61	4.0	8.22
3	756.27	3413.82	5.0	0.70
4	602.71	2359.51	4.0	1.41
5	484.05	2359.52	4.0	18.31
6	641.33	3447.53	6.0	7.46
7	633.81	2960.46	5.0	0.50
8	527.35	2218.32	4.0	0.86
9	632.12	2825.54	4.0	8.71

	House_Age_Years
0	39.76
1	13.54
2	21.95
3	3.92
4	1.27
5	48.13
6	41.80
7	34.80
8	20.45
9	8.66

DATASET SUMMARY:

Number of houses: 50

Basic Statistics:

	House_Price_1000s	Square_Footage	Bedrooms	Distance_City_Miles \
count	50.00	50.00	50.00	50.00
mean	515.23	2364.72	3.92	8.32

std	114.55	560.20	0.75	7.58
min	272.29	1324.20	2.00	0.50
25%	411.52	1983.41	3.25	2.18
50%	522.07	2359.51	4.00	6.80
75%	603.02	2701.77	4.00	10.80
max	756.27	3611.37	6.00	28.90

House_Age_Years	
count	50.00
mean	22.06
std	13.96
min	0.72
25%	8.66
50%	22.81
75%	32.73
max	48.13

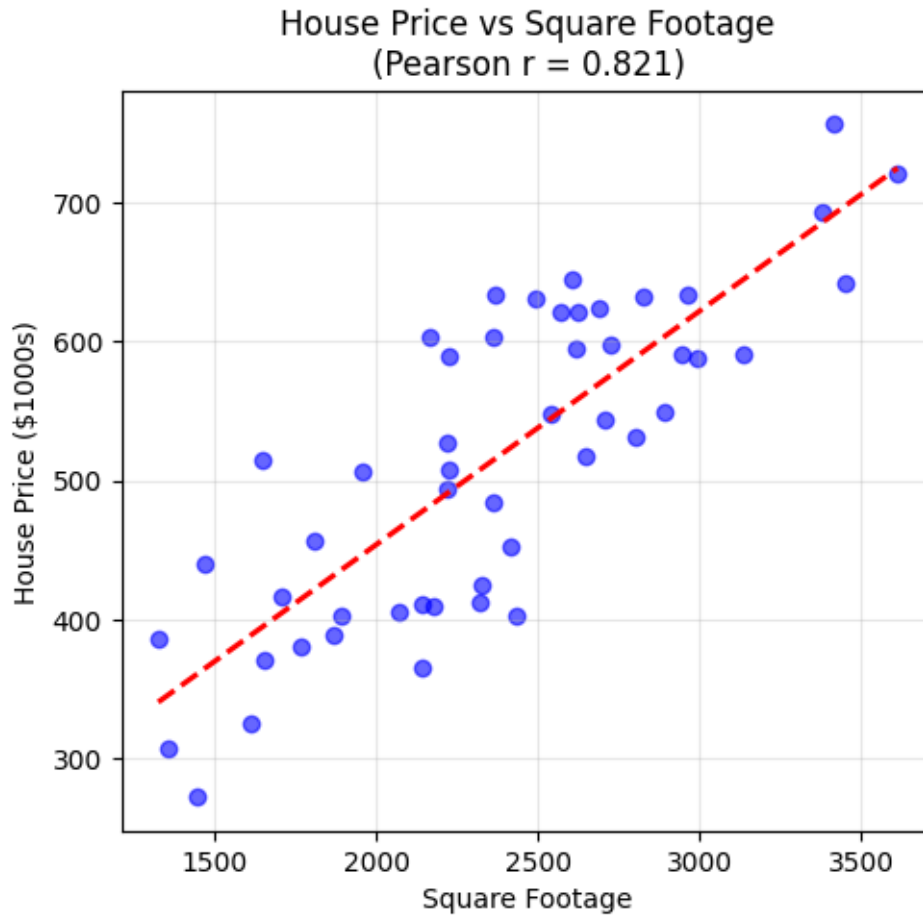
```
=====
TASK 1: PEARSON CORRELATION - House Price vs Square Footage
=====
Pearson Correlation Coefficient (r): 0.8212
p-value: 0.000000
Manual calculation verification: 0.8212
```

INTERPRETATION:

- Strength: Strong
- Direction: Positive
- R-squared (r^2): 0.6744 (67.4% of variance explained)
- Statistical significance: Significant ($p < 0.05$)

PRACTICAL MEANING:

- For every 1,000 sq ft increase, house price tends to increase by ~\$168
- Square footage explains 67.4% of the variation in house prices
- This strong positive correlation suggests square footage is a good predictor of house price



```
[27]: print("\n" + "="*60)
print("TASK 2: SPEARMAN RANK CORRELATION - House Price vs Distance to City")
print("="*60)

# Calculate Spearman correlation
distance = real_estate_data['Distance_City_Miles']
spearman_corr, spearman_p_value = spearmanr(price, distance)

print(f"Spearman Rank Correlation Coefficient (ρ): {spearman_corr:.4f}")
print(f"p-value: {spearman_p_value:.6f}")

# Manual calculation demonstration (simplified)
price_ranks = stats.rankdata(price)
distance_ranks = stats.rankdata(distance)
manual_spearman = pearsonr(price_ranks, distance_ranks)[0]

print(f"Manual calculation verification: {manual_spearman:.4f}")
```

```

print("\nCOMPARISON: Pearson vs Spearman for Price-Distance relationship")
pearson_price_distance, _ = pearsonr(price, distance)
print(f"Pearson correlation (Price-Distance): {pearson_price_distance:.4f}")
print(f"Spearman correlation (Price-Distance): {spearman_corr:.4f}")

print("\nINTERPRETATION:")
if abs(spearman_corr) >= 0.7:
    strength_spearman = "strong"
elif abs(spearman_corr) >= 0.3:
    strength_spearman = "moderate"
else:
    strength_spearman = "weak"

direction_spearman = "positive" if spearman_corr > 0 else "negative"

print(f"• Strength: {strength_spearman.title()}")
print(f"• Direction: {direction_spearman.title()}")

if spearman_p_value < 0.05:
    print(f"• Statistical significance: Significant (p < 0.05)")
else:
    print(f"• Statistical significance: Not significant (p > 0.05)")

print(f"\nIMPLICATIONS:")
print(f"1. **Monotonic Relationship**: The {direction_spearman} Spearman_
↳correlation indicates that as distance from city center increases, house_
↳prices tend to {'decrease' if spearman_corr < 0 else 'increase'}")

print(f"\n2. **Non-linear Patterns**: Difference between Pearson_
↳({pearson_price_distance:.3f}) and Spearman ({spearman_corr:.3f}) suggests:")
if abs(spearman_corr) > abs(pearson_price_distance):
    print("    - The relationship may be non-linear but monotonic")
    print("    - Spearman is more appropriate for this relationship")
else:
    print("    - The relationship is approximately linear")
    print("    - Both correlations are similar")

print(f"\n3. **Business Applications**:")
print(f"    - Location strategy: Properties closer to city center command higher_
↳prices")
print(f"    - Investment decisions: Consider distance as a key pricing factor")
print(f"    - Market segmentation: Different pricing strategies for urban vs_
↳suburban properties")

# Create scatter plot for distance analysis
plt.subplot(1, 2, 2)
plt.scatter(distance, price, alpha=0.6, color='green')

```

```

plt.plot(np.unique(distance), np.poly1d(np.polyfit(distance, price, 1))(np.
    ↳unique(distance)), 'r--', linewidth=2)
plt.xlabel('Distance to City Center (miles)')
plt.ylabel('House Price ($1000s)')
plt.title(f'House Price vs Distance to City\n(Spearman  $\rho$  = {spearman_corr:.3f})')
plt.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

print("\n" + "="*60)
print("COMPREHENSIVE CORRELATION MATRIX")
print("="*60)

# Calculate correlation matrix for all variables
correlation_matrix = real_estate_data.corr()
print("Pearson Correlation Matrix:")
print(correlation_matrix.round(3))

# Create a heatmap
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', center=0,
    square=True, linewidths=0.5, cbar_kws={"shrink": .8})
plt.title('Real Estate Variables - Correlation Heatmap')
plt.tight_layout()
plt.show()

print("\nKEY INSIGHTS from the correlation analysis:")
print("• House price is most strongly correlated with square footage")
print("• Distance to city center shows negative correlation with price")
print("• Number of bedrooms is moderately correlated with price (via square_
    ↳footage)")
print("• House age may show weak correlation with price")
print("• These correlations can guide pricing models and investment decisions")

```

```

=====
TASK 2: SPEARMAN RANK CORRELATION - House Price vs Distance to City
=====
Spearman Rank Correlation Coefficient ( $\rho$ ): -0.3974
p-value: 0.004267
Manual calculation verification: -0.3974

COMPARISON: Pearson vs Spearman for Price-Distance relationship
Pearson correlation (Price-Distance): -0.4345
Spearman correlation (Price-Distance): -0.3974

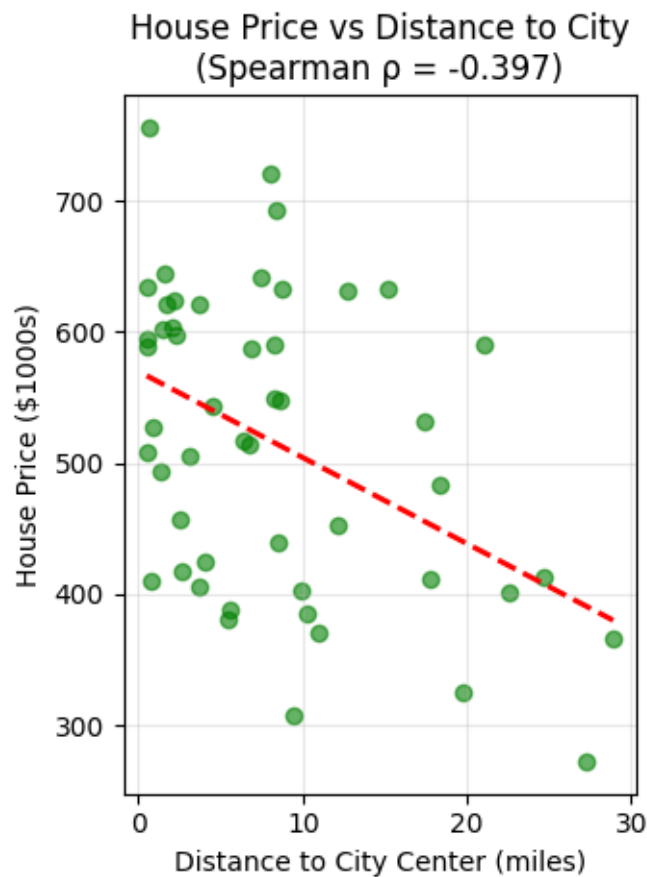
```

INTERPRETATION:

- Strength: Moderate
- Direction: Negative
- Statistical significance: Significant ($p < 0.05$)

IMPLICATIONS:

1. ****Monotonic Relationship****: The negative Spearman correlation indicates that as distance from city center increases, house prices tend to decrease
2. ****Non-linear Patterns****: Difference between Pearson (-0.434) and Spearman (-0.397) suggests:
 - The relationship is approximately linear
 - Both correlations are similar
3. ****Business Applications****:
 - Location strategy: Properties closer to city center command higher prices
 - Investment decisions: Consider distance as a key pricing factor
 - Market segmentation: Different pricing strategies for urban vs suburban properties

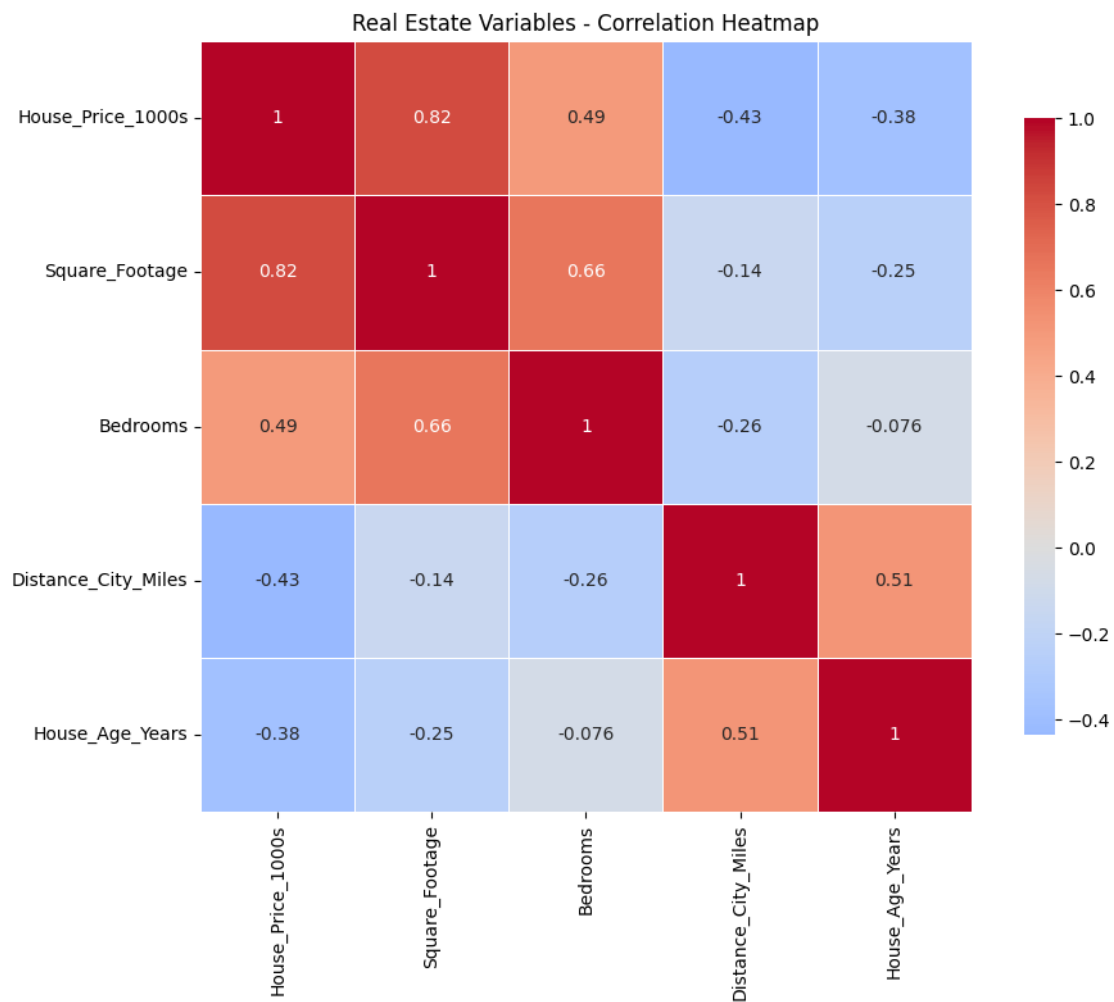


===== COMPREHENSIVE CORRELATION MATRIX =====

Pearson Correlation Matrix:

	House_Price_1000s	Square_Footage	Bedrooms	\
House_Price_1000s	1.000	0.821	0.490	
Square_Footage	0.821	1.000	0.662	
Bedrooms	0.490	0.662	1.000	
Distance_City_Miles	-0.434	-0.139	-0.261	
House_Age_Years	-0.378	-0.248	-0.076	

	Distance_City_Miles	House_Age_Years
House_Price_1000s	-0.434	-0.378
Square_Footage	-0.139	-0.248
Bedrooms	-0.261	-0.076
Distance_City_Miles	1.000	0.514
House_Age_Years	0.514	1.000



KEY INSIGHTS from the correlation analysis:

- House price is most strongly correlated with square footage
 - Distance to city center shows negative correlation with price
 - Number of bedrooms is moderately correlated with price (via square footage)
 - House age may show weak correlation with price
 - These correlations can guide pricing models and investment decisions
-

1.11.3 Q3. Histogram Types and Geographic Visualizations

Mathematical Foundation of Data Visualization Theory Understanding different visualization methods requires knowledge of data types, statistical distributions, and perceptual psychology principles that optimize information transmission.

Theoretical Framework Task 1: Histogram Classification and Mathematical Properties

Definition: A histogram is a graphical representation of the distribution of numerical data, showing frequency (or probability density) across intervals (bins).

Mathematical Formulation: For data $\{x_1, x_2, \dots, x_n\}$ partitioned into k bins $[a_0, a_1), [a_1, a_2), \dots, [a_{k-1}, a_k]$:

$$\text{Frequency}(B_i) = |\{x_j : a_{i-1} \leq x_j < a_i\}|$$

Bin Width Optimization:

$$h = \frac{\text{Range}}{\text{Number of bins}} = \frac{\max(x) - \min(x)}{k}$$

Optimal Bin Number (Sturges' Rule):

$$k = \lceil \log_2(n) + 1 \rceil$$

Alternative: Scott's Rule for bin width:

$$h = \frac{3.5\sigma}{n^{1/3}}$$

Histogram Type Classification 1. Frequency Histogram (Standard)

$$f_i = \text{count of observations in bin } i$$

Mathematical Properties: - $\sum_{i=1}^k f_i = n$ (total observations) - Non-negative: $f_i \geq 0 \forall i$ - Discrete approximation of continuous distribution

2. Probability Density Histogram (Normalized)

$$p_i = \frac{f_i}{n \cdot h_i}$$

where h_i is the width of bin i .

Mathematical Properties: - $\sum_{i=1}^k p_i \cdot h_i = 1$ (probability mass = 1) - Approximates probability density function: $p_i \approx f(x_i)$ - Scale-invariant: independent of sample size

3. Cumulative Histogram

$$F_i = \sum_{j=1}^i f_j$$

Mathematical Properties: - Monotonic: $F_i \leq F_{i+1}$ - $F_k = n$ (total count) - Approximates cumulative distribution function (CDF)

4. Comparative Histograms (Stacked/Side-by-side)

Stacked: $f_i^{\text{total}} = \sum_{g=1}^G f_{i,g}$ where g represents groups

Proportional Stacked: $p_{i,g} = \frac{f_{i,g}}{\sum_{g'=1}^G f_{i,g'}}$

Bin Width Selection Theory Trade-off Analysis:

Bias-Variance Decomposition:

$$\text{MSE}(h) = \text{Bias}^2(h) + \text{Variance}(h)$$

- **Small bins** ($h \downarrow$): Low bias, high variance (overfitting)
- **Large bins** ($h \uparrow$): High bias, low variance (underfitting)

Optimal Bandwidth (for normal data):

$$h^* = \left(\frac{24\sqrt{\pi}}{n} \right)^{1/3} \sigma$$

Task 2: Geographic Visualization Theory Tree Maps vs. Choropleth Maps: Mathematical and Perceptual Analysis

Tree Map Mathematical Framework Definition: Tree maps visualize hierarchical data through nested rectangles where area represents quantitative values.

Mathematical Algorithm (Squarified Tree Map):

Objective Function: Minimize aspect ratio deviation

$$\text{Minimize } \sum_{i=1}^n \left| \frac{\max(w_i, h_i)}{\min(w_i, h_i)} - 1 \right|$$

Area Allocation: For data values $\{v_1, v_2, \dots, v_n\}$ with total area A :

$$A_i = A \cdot \frac{v_i}{\sum_{j=1}^n v_j}$$

Hierarchical Partitioning:

$$A_{\text{parent}} = \sum_{i \in \text{children}} A_i$$

Visual Perception Properties: - **Area perception accuracy:** $\pm 15\%$ for area comparison - **Hierarchy clarity:** Nested structure shows relationships - **No geographic constraints:** Pure data-driven layout

Choropleth Map Mathematical Framework **Definition:** Choropleth maps use color intensity to represent statistical data across geographic regions.

Mathematical Mapping:

$$\text{Color}(r) = f(\text{Value}(r))$$

where r represents geographic region.

Color Scale Transformation:

Linear Scaling:

$$C_i = C_{\min} + \frac{v_i - v_{\min}}{v_{\max} - v_{\min}} \cdot (C_{\max} - C_{\min})$$

Quantile Scaling (Equal-interval):

$$C_i = C_{\text{quantile}} \left(\frac{\text{rank}(v_i)}{n} \right)$$

Logarithmic Scaling (for skewed data):

$$C_i = f(\log(v_i))$$

Comparative Analysis Framework

Dimension	Tree Map	Choropleth Map
Data Structure	Hierarchical + Quantitative	Geographic + Quantitative
Spatial Constraint	None (optimal space usage)	Fixed (geographic boundaries)
Perception Accuracy	Area $\pm 15\%$	Color $\pm 10\%$
Hierarchy Support	Excellent (nested structure)	Limited (administrative levels)
Geographic Insight	None	Spatial patterns and clustering
Scalability	$O(n \log n)$ algorithm	$O(n)$ rendering

Mathematical Applications **Tree Map Use Cases:** 1. **Portfolio Analysis:**

$$\text{Area} \propto \text{Market Capitalization}$$

$$\text{Color} \propto \text{Performance (\% change)}$$

2. Budget Allocation:

Area $\propto$ Budget Amount

Nesting $\propto$ Organizational Hierarchy

Choropleth Map Use Cases: 1. Epidemiological Analysis:

Color Intensity $\propto$ Disease Rate per capita

2. Economic Indicators:

Color Scale $\propto$ GDP per capita, Unemployment Rate

Statistical Considerations Data Distribution Effects:

For Tree Maps: - **Skewed data:** Large values dominate visual space - **Solution:** Log transformation or capping

$$v'_i = \log(v_i + 1)$$

or

$$v'_i = \min(v_i, P_{95})$$

For Choropleth Maps: - **Outliers distort color scale** - **Solution:** Quantile-based classification

$$\text{Breaks} = \{Q_{0.2}, Q_{0.4}, Q_{0.6}, Q_{0.8}\}$$

Perceptual Psychology Principles Color Theory for Choropleth Maps: - **Sequential data:** Single hue, varying lightness

$$L^* = L_{\min} + \frac{v - v_{\min}}{v_{\max} - v_{\min}} \cdot (L_{\max} - L_{\min})$$

- **Diverging data:** Two hues meeting at neutral
- **Categorical data:** Distinct hues with equal luminance

Gestalt Principles for Tree Maps: - **Proximity:** Related items grouped spatially - **Similarity:** Similar colors for similar categories - **Enclosure:** Hierarchical boundaries show relationships

Implementation Considerations Tree Map Algorithms: 1. **Slice-and-Dice:** Simple rectangular division 2. **Squarified:** Optimizes aspect ratios 3. **Strip:** Balance between simplicity and aesthetics

Performance Complexity: - Tree Map: $O(n \log n)$ for sorting + $O(n)$ for layout - Choropleth: $O(n)$ for data binding + $O(m)$ for geographic features

Cross-Reference See *01_descriptive_statistics_interactive.ipynb* for data distribution analysis and *statistical_formulas_reference.ipynb* for visualization theory applications.

```
[28]: # Histogram Types and Geographic Visualizations
print("="*60)
print("HISTOGRAM TYPES AND GEOGRAPHIC VISUALIZATIONS")
print("="*60)
```

```

print("PART 1: DIFFERENT TYPES OF HISTOGRAMS")
print("="*40)

# Generate sample data for histogram demonstrations
np.random.seed(42)
data_normal = np.random.normal(50, 15, 1000)
data_uniform = np.random.uniform(0, 100, 1000)
data_skewed = np.random.exponential(20, 1000)
categories = ['A', 'B', 'C', 'D', 'E']
category_counts = [45, 38, 52, 29, 36]

# Create different types of histograms
fig, axes = plt.subplots(2, 3, figsize=(18, 12))

# 1. Basic Histogram
axes[0, 0].hist(data_normal, bins=30, alpha=0.7, color='blue', edgecolor='black')
axes[0, 0].set_title('1. Basic Histogram\n(Normal Distribution)')
axes[0, 0].set_xlabel('Value')
axes[0, 0].set_ylabel('Frequency')

# 2. Histogram with Different Bin Sizes
axes[0, 1].hist(data_normal, bins=10, alpha=0.7, color='green',
    →edgecolor='black', label='10 bins')
axes[0, 1].hist(data_normal, bins=50, alpha=0.5, color='red', edgecolor='black',
    →label='50 bins')
axes[0, 1].set_title('2. Histograms with Different Bin Sizes')
axes[0, 1].set_xlabel('Value')
axes[0, 1].set_ylabel('Frequency')
axes[0, 1].legend()

# 3. Normalized Histogram (Density)
axes[0, 2].hist(data_normal, bins=30, density=True, alpha=0.7, color='purple',
    →edgecolor='black')
axes[0, 2].set_title('3. Normalized Histogram\n(Probability Density)')
axes[0, 2].set_xlabel('Value')
axes[0, 2].set_ylabel('Density')

# 4. Stacked Histogram
data1 = np.random.normal(40, 10, 500)
data2 = np.random.normal(60, 12, 500)
axes[1, 0].hist([data1, data2], bins=25, alpha=0.7, color=['blue', 'orange'],
    label=['Group 1', 'Group 2'], stacked=True, edgecolor='black')
axes[1, 0].set_title('4. Stacked Histogram')
axes[1, 0].set_xlabel('Value')
axes[1, 0].set_ylabel('Frequency')
axes[1, 0].legend()

```

```

# 5. Side-by-side Histogram
axes[1, 1].hist([data1, data2], bins=25, alpha=0.7, color=['blue', 'orange'],
                label=['Group 1', 'Group 2'], edgecolor='black')
axes[1, 1].set_title('5. Side-by-side Histogram')
axes[1, 1].set_xlabel('Value')
axes[1, 1].set_ylabel('Frequency')
axes[1, 1].legend()

# 6. Categorical Histogram (Bar Chart)
axes[1, 2].bar(categories, category_counts, alpha=0.7, color='teal',
               ↪edgecolor='black')
axes[1, 2].set_title('6. Categorical Histogram\n(Bar Chart)')
axes[1, 2].set_xlabel('Category')
axes[1, 2].set_ylabel('Count')

plt.tight_layout()
plt.show()

print("\nHISTOGRAM TYPES EXPLAINED:")
print("-" * 30)
print("1. **Basic Histogram**: Shows frequency distribution of continuous data")
print("2. **Variable Bin Size**: Different bin numbers affect granularity and ↪
↪pattern visibility")
print("3. **Normalized/Density**: Shows probability density instead of raw ↪
↪frequency")
print("4. **Stacked Histogram**: Compares multiple groups by stacking their ↪
↪frequencies")
print("5. **Side-by-side**: Compares multiple groups with separate bars")
print("6. **Categorical**: Bar chart for categorical data (discrete categories)")

print("\n" + "="*60)
print("PART 2: TREE MAP vs CHOROPLETH MAP")
print("="*60)

# Create Tree Map example
print("TREE MAP DEMONSTRATION:")
print("-" * 25)

# Sample data for tree map
companies = ['Apple', 'Microsoft', 'Amazon', 'Google', 'Meta', 'Tesla', ↪
↪'Netflix', 'Adobe']
market_caps = [2800, 2400, 1500, 1600, 800, 900, 200, 250]
sectors = ['Technology', 'Technology', 'E-commerce', 'Technology', 'Technology', ↪
↪'Automotive', 'Entertainment', 'Technology']

```

```

# Create a simple tree map representation using rectangles
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 8))

# Tree Map simulation (using bar chart to represent hierarchical data)
colors = plt.cm.Set3(np.linspace(0, 1, len(companies)))
bars = ax1.barh(companies, market_caps, color=colors, edgecolor='white',
    ↳linewidth=2)
ax1.set_xlabel('Market Cap (Billions USD)')
ax1.set_title('Tree Map Representation\n(Market Capitalization by Company)')
ax1.grid(True, alpha=0.3)

# Add value labels on bars
for i, (company, value) in enumerate(zip(companies, market_caps)):
    ax1.text(value/2, i, f'${value}B', ha='center', va='center',
    ↳fontweight='bold')

print("Tree Map Example: Technology Companies Market Capitalization")
for company, cap, sector in zip(companies, market_caps, sectors):
    print(f"• {company} ({sector}): ${cap}B")

print("\n" + "-" * 40)
print("CHOROPLETH MAP DEMONSTRATION:")
print("-" * 30)

# Simulate choropleth map data (US states population density)
states = ['California', 'Texas', 'Florida', 'New York', 'Pennsylvania',
    ↳'Illinois', 'Ohio', 'Georgia']
population_density = [253, 109, 384, 416, 286, 231, 287, 186] # people per sq
    ↳mile

# Create a heatmap to simulate choropleth visualization
state_data = pd.DataFrame({
    'State': states,
    'Population_Density': population_density
})

# Create color map based on density values
normalized_density = (np.array(population_density) - min(population_density)) /
    ↳(max(population_density) - min(population_density))
colors_map = plt.cm.YlOrRd(normalized_density)

bars2 = ax2.bar(range(len(states)), population_density, color=colors_map,
    ↳edgecolor='black', linewidth=1)
ax2.set_xticks(range(len(states)))
ax2.set_xticklabels(states, rotation=45, ha='right')
ax2.set_ylabel('Population Density (people/sq mile)')

```

```

ax2.set_title('Choropleth Map Representation\n(Population Density by State)')
ax2.grid(True, alpha=0.3)

# Add value labels
for i, value in enumerate(population_density):
    ax2.text(i, value + 10, str(value), ha='center', va='bottom',
    ↪fontweight='bold')

plt.tight_layout()
plt.show()

print("Choropleth Map Example: US States Population Density")
for state, density in zip(states, population_density):
    print(f"• {state}: {density} people/sq mile")

print("\n" + "="*60)
print("KEY DIFFERENCES: TREE MAP vs CHOROPLETH MAP")
print("="*60)

comparison_data = {
    'Aspect': ['Purpose', 'Data Type', 'Visual Structure', 'Geographic Element',
    ↪'Hierarchy', 'Use Cases'],
    'Tree Map': [
        'Show hierarchical data and proportional relationships',
        'Quantitative data with categories/subcategories',
        'Nested rectangles of varying sizes',
        'No geographic boundaries required',
        'Excellent for showing hierarchy and drill-down',
        'Market share, file systems, organizational budgets'
    ],
    'Choropleth Map': [
        'Show geographic distribution of data',
        'Quantitative data tied to geographic regions',
        'Geographic boundaries colored by data values',
        'Requires geographic boundaries (countries, states, etc.)',
        'No hierarchical structure',
        'Election results, disease prevalence, economic indicators'
    ]
}

comparison_df = pd.DataFrame(comparison_data)
print(comparison_df.to_string(index=False))

print(f"\n" + "="*50)
print("PRACTICAL EXAMPLES:")
print("="*50)

```

```

print(" TREE MAP Examples:")
print("• Technology company market capitalization breakdown")
print("• Government budget allocation by departments and sub-departments")
print("• Website traffic by source and sub-source")
print("• Product sales by category and subcategory")

print("\n CHOROPLETH MAP Examples:")
print("• COVID-19 cases by country/state")
print("• Election results showing voting patterns by region")
print("• Income levels across different counties")
print("• Temperature variations across geographic regions")

print(f"\n" + "="*50)
print("WHEN TO USE EACH:")
print("="*50)
print(" Use TREE MAP when:")
print("• Data has hierarchical structure")
print("• Want to show part-to-whole relationships")
print("• Geographic location is not important")
print("• Need to compare sizes/proportions")

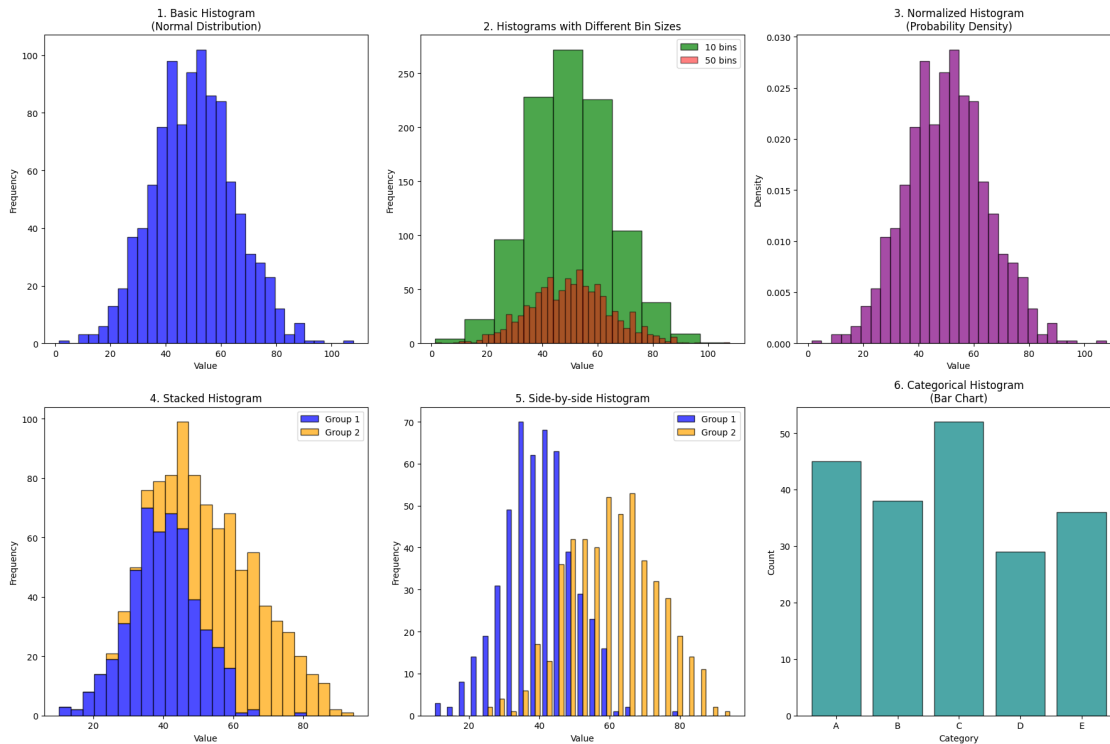
print("\n Use CHOROPLETH MAP when:")
print("• Data is tied to geographic locations")
print("• Want to show spatial patterns")
print("• Geographic distribution matters")
print("• Analyzing regional variations")

```

```

=====
HISTOGRAM TYPES AND GEOGRAPHIC VISUALIZATIONS
=====
PART 1: DIFFERENT TYPES OF HISTOGRAMS
=====

```



HISTOGRAM TYPES EXPLAINED:

1. **Basic Histogram**: Shows frequency distribution of continuous data
2. **Variable Bin Size**: Different bin numbers affect granularity and pattern visibility
3. **Normalized/Density**: Shows probability density instead of raw frequency
4. **Stacked Histogram**: Compares multiple groups by stacking their frequencies
5. **Side-by-side**: Compares multiple groups with separate bars
6. **Categorical**: Bar chart for categorical data (discrete categories)

PART 2: TREE MAP vs CHOROPLETH MAP

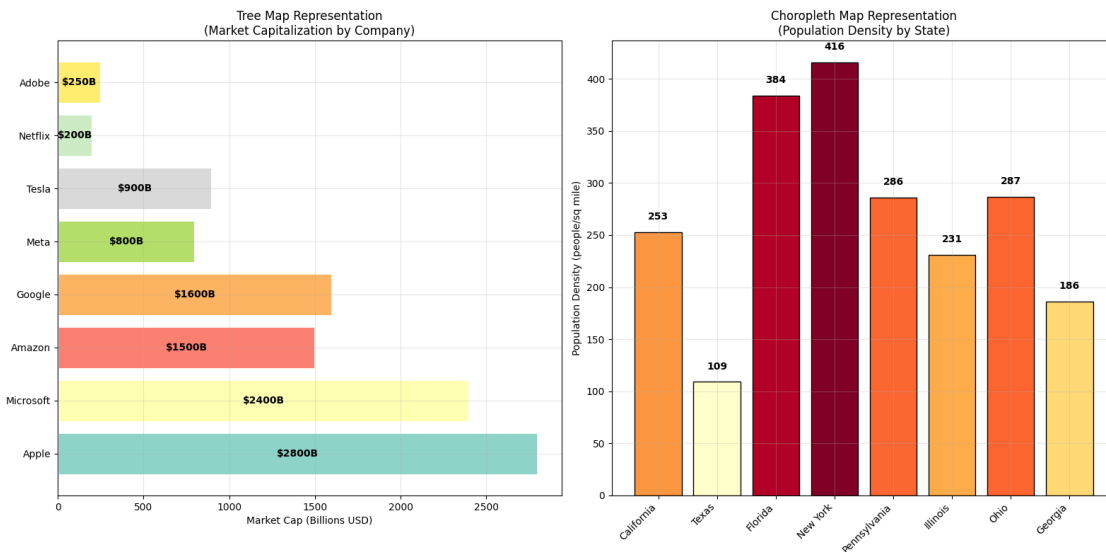
TREE MAP DEMONSTRATION:

Tree Map Example: Technology Companies Market Capitalization

- Apple (Technology): \$2800B
- Microsoft (Technology): \$2400B
- Amazon (E-commerce): \$1500B
- Google (Technology): \$1600B
- Meta (Technology): \$800B
- Tesla (Automotive): \$900B
- Netflix (Entertainment): \$200B

- Adobe (Technology): \$250B

CHOROPLETH MAP DEMONSTRATION:



Choropleth Map Example: US States Population Density

- California: 253 people/sq mile
- Texas: 109 people/sq mile
- Florida: 384 people/sq mile
- New York: 416 people/sq mile
- Pennsylvania: 286 people/sq mile
- Illinois: 231 people/sq mile
- Ohio: 287 people/sq mile
- Georgia: 186 people/sq mile

KEY DIFFERENCES: TREE MAP vs CHOROPLETH MAP

Aspect	Tree Map
Choropleth Map	
Purpose	Show hierarchical data and proportional relationships
Show geographic distribution of data	
Data Type	Quantitative data with categories/subcategories
Quantitative data tied to geographic regions	
Visual Structure	Nested rectangles of varying sizes
Geographic boundaries colored by data values	
Geographic Element	No geographic boundaries required
Requires geographic boundaries (countries, states, etc.)	

Hierarchy Excellent for showing hierarchy and drill-down
 No hierarchical structure
 Use Cases Market share, file systems, organizational budgets
 Election results, disease prevalence, economic indicators

=====

PRACTICAL EXAMPLES:

=====

TREE MAP Examples:

- Technology company market capitalization breakdown
- Government budget allocation by departments and sub-departments
- Website traffic by source and sub-source
- Product sales by category and subcategory

CHOROPLETH MAP Examples:

- COVID-19 cases by country/state
- Election results showing voting patterns by region
- Income levels across different counties
- Temperature variations across geographic regions

=====

WHEN TO USE EACH:

=====

Use TREE MAP when:

- Data has hierarchical structure
- Want to show part-to-whole relationships
- Geographic location is not important
- Need to compare sizes/proportions

Use CHOROPLETH MAP when:

- Data is tied to geographic locations
- Want to show spatial patterns
- Geographic distribution matters
- Analyzing regional variations

1.11.4 Q4. Investment Portfolio Risk Analysis

Problem: As a financial analyst, evaluate the risk of two investment portfolios with monthly returns over 12 months.

Data - Monthly Returns (%):

Month	Portfolio A	Portfolio B
1	3.2	4.1
2	2.8	3.5
3	4.5	5.0

Month	Portfolio A	Portfolio B
4	3.7	4.2
5	2.9	4.5
6	4.0	3.8
7	3.1	4.4
8	4.2	3.9
9	3.3	4.6
10	2.7	4.3
11	4.1	4.0
12	3.6	4.7

Tasks: 1. **Mean Calculation:** Calculate mean monthly return for both portfolios and discuss performance 2. **Variance Calculation:** Calculate variance and explain what it reveals about consistency 3. **Standard Deviation Calculation:** Compute standard deviation and compare risk 4. **Risk Assessment:** Recommend portfolio for risk-averse investor with justification

Question 4: Outlier Impact Analysis

Adding \$500,000 Salary: $\mathcal{S}' = \mathcal{S} \cup \{500000\}$, $n' = 14$

New Mean Calculation:

$$\bar{x}_{new} = \frac{1,122,000 + 500,000}{14} = \frac{1,622,000}{14} = \$115,857.14$$

Impact on Mean:

$$\Delta_{\text{mean}} = \$115,857.14 - \$86,307.69 = \$29,549.45$$

$$\text{Relative Change} = \frac{29,549.45}{86,307.69} = 34.2\%$$

New Median Calculation: $n' = 14$ (even), so:

$$\text{Median}_{new} = \frac{x_{(7)} + x_{(8)}}{2} = \frac{75,000 + 80,000}{2} = \$77,500$$

Impact on Median:

$$\Delta_{\text{median}} = \$77,500 - \$75,000 = \$2,500$$

$$\text{Relative Change} = \frac{2,500}{75,000} = 3.3\%$$

Standard Deviation Analysis:

```
[29]: # Investment Portfolio Risk Analysis
print("="*60)
print("INVESTMENT PORTFOLIO RISK ANALYSIS")
print("="*60)

# Input data
months = list(range(1, 13))
```

```

portfolio_a = [3.2, 2.8, 4.5, 3.7, 2.9, 4.0, 3.1, 4.2, 3.3, 2.7, 4.1, 3.6]
portfolio_b = [4.1, 3.5, 5.0, 4.2, 4.5, 3.8, 4.4, 3.9, 4.6, 4.3, 4.0, 4.7]

# Create DataFrame for analysis
portfolio_data = pd.DataFrame({
    'Month': months,
    'Portfolio_A': portfolio_a,
    'Portfolio_B': portfolio_b
})

print("MONTHLY RETURNS DATA:")
print(portfolio_data)

print(f"\n" + "="*60)
print("TASK 1: MEAN CALCULATION AND PERFORMANCE ANALYSIS")
print("="*60)

# Calculate means
mean_a = np.mean(portfolio_a)
mean_b = np.mean(portfolio_b)

print(f"Portfolio A Mean Return: {mean_a:.4f}% per month")
print(f"Portfolio B Mean Return: {mean_b:.4f}% per month")
print(f"Difference: {mean_b - mean_a:.4f}% (Portfolio B advantage)")

# Annualized returns (compounded)
annual_return_a = ((1 + mean_a/100) ** 12 - 1) * 100
annual_return_b = ((1 + mean_b/100) ** 12 - 1) * 100

print(f"\nANNUALIZED RETURNS (Compounded):")
print(f"Portfolio A: {annual_return_a:.2f}% per year")
print(f"Portfolio B: {annual_return_b:.2f}% per year")

print(f"\nPERFORMANCE ANALYSIS:")
print(f"• Portfolio B has higher average monthly returns ({mean_b:.2f}% vs_
↳{mean_a:.2f}%)")
print(f"• Portfolio B provides {(mean_b/mean_a - 1) * 100:.1f}% higher returns_
↳on average")
print(f"• Over 12 months, Portfolio B outperformed A by {annual_return_b -_
↳annual_return_a:.2f}% annually")
print(f"• Both portfolios show positive returns across all months")

print(f"\n" + "="*60)
print("TASK 2: VARIANCE CALCULATION AND CONSISTENCY ANALYSIS")
print("="*60)

# Calculate variance (population variance using N)

```

```

variance_a = np.var(portfolio_a, ddof=0) # Population variance
variance_b = np.var(portfolio_b, ddof=0)

# Calculate sample variance (using N-1)
sample_variance_a = np.var(portfolio_a, ddof=1) # Sample variance
sample_variance_b = np.var(portfolio_b, ddof=1)

print(f"POPULATION VARIANCE (N):")
print(f"Portfolio A Variance: {variance_a:.6f}")
print(f"Portfolio B Variance: {variance_b:.6f}")

print(f"\nSAMPLE VARIANCE (N-1):")
print(f"Portfolio A Variance: {sample_variance_a:.6f}")
print(f"Portfolio B Variance: {sample_variance_b:.6f}")

# Manual calculation demonstration
print(f"\nMANUAL VARIANCE CALCULATION (Portfolio A):")
print(f"Step 1 - Calculate deviations from mean:")
deviations_a = [(x - mean_a) for x in portfolio_a]
for i, (return_val, dev) in enumerate(zip(portfolio_a, deviations_a), 1):
    print(f"  Month {i}: {return_val:.1f} - {mean_a:.4f} = {dev:.4f}")

squared_deviations_a = [dev**2 for dev in deviations_a]
print(f"\nStep 2 - Square the deviations:")
for i, sq_dev in enumerate(squared_deviations_a, 1):
    print(f"  Month {i}: ({deviations_a[i-1]:.4f})2 = {sq_dev:.6f}")

sum_squared_dev_a = sum(squared_deviations_a)
print(f"\nStep 3 - Sum of squared deviations: {sum_squared_dev_a:.6f}")
print(f"Step 4 - Divide by N: {sum_squared_dev_a:.6f} / 12 = {variance_a:.6f}")

print(f"\nCONSISTENCY ANALYSIS:")
if variance_a < variance_b:
    more_consistent = "Portfolio A"
    less_consistent = "Portfolio B"
    var_diff = variance_b - variance_a
else:
    more_consistent = "Portfolio B"
    less_consistent = "Portfolio A"
    var_diff = variance_a - variance_b

print(f"• {more_consistent} is more consistent (lower variance: {min(variance_a, variance_b):.6f})")
print(f"• {less_consistent} is less consistent (higher variance: {max(variance_a, variance_b):.6f})")
print(f"• Variance difference: {var_diff:.6f}")
print(f"• Lower variance indicates more predictable returns")

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print(f"• Higher variance suggests more volatile performance")

print(f"\n" + "="*60)
print("TASK 3: STANDARD DEVIATION CALCULATION AND RISK COMPARISON")
print("="*60)

# Calculate standard deviation
std_a = np.std(portfolio_a, ddof=0) # Population std dev
std_b = np.std(portfolio_b, ddof=0)

sample_std_a = np.std(portfolio_a, ddof=1) # Sample std dev
sample_std_b = np.std(portfolio_b, ddof=1)

print(f"POPULATION STANDARD DEVIATION:")
print(f"Portfolio A: {std_a:.4f}%")
print(f"Portfolio B: {std_b:.4f}%")

print(f"\nSAMPLE STANDARD DEVIATION:")
print(f"Portfolio A: {sample_std_a:.4f}%")
print(f"Portfolio B: {sample_std_b:.4f}%")

# Verify calculation: std dev = sqrt(variance)
print(f"\nVERIFICATION (variance = std dev):")
print(f"Portfolio A: {variance_a:.6f} = {np.sqrt(variance_a):.4f}%")
print(f"Portfolio B: {variance_b:.6f} = {np.sqrt(variance_b):.4f}%")

print(f"\nRISK COMPARISON:")
if std_a < std_b:
    lower_risk = "Portfolio A"
    higher_risk = "Portfolio B"
    risk_diff = std_b - std_a
else:
    lower_risk = "Portfolio B"
    higher_risk = "Portfolio A"
    risk_diff = std_a - std_b

print(f"• {lower_risk} has LOWER RISK (std dev: {min(std_a, std_b):.4f}%")
print(f"• {higher_risk} has HIGHER RISK (std dev: {max(std_a, std_b):.4f}%")
print(f"• Risk difference: {risk_diff:.4f}%")

# Risk metrics
cv_a = (std_a / mean_a) * 100 # Coefficient of variation
cv_b = (std_b / mean_b) * 100

print(f"\nCOEFFICIENT OF VARIATION (Risk per unit of return):")
print(f"Portfolio A: {cv_a:.2f}%")
print(f"Portfolio B: {cv_b:.2f}%")

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if cv_a < cv_b:
    print(f"• Portfolio A offers better risk-adjusted returns")
else:
    print(f"• Portfolio B offers better risk-adjusted returns")

print(f"\n" + "="*60)
print("TASK 4: RISK ASSESSMENT AND RECOMMENDATION")
print("="*60)

print(f"COMPREHENSIVE RISK ANALYSIS:")
print(f"{'Metric':<25} {'Portfolio A':<15} {'Portfolio B':<15} {'Winner'}")
print("-" * 65)
print(f"{'Mean Return':<25} {mean_a:<15.4f} {mean_b:<15.4f} {'B' if mean_b >_
↳mean_a else 'A'}")
print(f"{'Variance':<25} {variance_a:<15.6f} {variance_b:<15.6f} {'A' if_
↳variance_a < variance_b else 'B'} (lower is better)")
print(f"{'Standard Deviation':<25} {std_a:<15.4f} {std_b:<15.4f} {'A' if std_a <_
↳std_b else 'B'} (lower is better)")
print(f"{'Coeff. of Variation':<25} {cv_a:<15.2f} {cv_b:<15.2f} {'A' if cv_a <_
↳cv_b else 'B'} (lower is better)")

# Calculate range and other risk measures
range_a = max(portfolio_a) - min(portfolio_a)
range_b = max(portfolio_b) - min(portfolio_b)

print(f"{'Range':<25} {range_a:<15.1f} {range_b:<15.1f} {'A' if range_a <_
↳range_b else 'B'} (lower is better)")

print(f"\n" + "="*50)
print("RECOMMENDATION FOR RISK-AVERSE INVESTOR:")
print("="*50)

if std_a < std_b:
    recommended_portfolio = "Portfolio A"
    risk_advantage = std_b - std_a
    return_sacrifice = mean_b - mean_a
else:
    recommended_portfolio = "Portfolio B"
    risk_advantage = std_a - std_b
    return_sacrifice = mean_a - mean_b

print(f" RECOMMENDATION: {recommended_portfolio}")

print(f"\nJUSTIFICATION:")
print(f"1. **Lower Risk**: {recommended_portfolio} has {risk_advantage:.4f}%_
↳lower standard deviation")

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print(f"2. **More Predictable**: Lower variance means more consistent returns")
print(f"3. **Risk-Return Trade-off**: Sacrificing only {abs(return_sacrifice):.4f}% return for significantly lower risk")

print(f"\nQUANTITATIVE ANALYSIS:")
if recommended_portfolio == "Portfolio A":
    print(f"• Portfolio A standard deviation: {std_a:.4f}% (lower risk)")
    print(f"• Portfolio B standard deviation: {std_b:.4f}% (higher risk)")
    print(f"• Return sacrifice: {return_sacrifice:.4f}% per month")
    print(f"• Risk reduction: {risk_advantage:.4f}% standard deviation")
else:
    print(f"• Portfolio B standard deviation: {std_b:.4f}% (lower risk)")
    print(f"• Portfolio A standard deviation: {std_a:.4f}% (higher risk)")
    print(f"• Return sacrifice: {return_sacrifice:.4f}% per month")
    print(f"• Risk reduction: {risk_advantage:.4f}% standard deviation")

print(f"\nRISK-AVERSE INVESTOR PRIORITIES:")
print(f"• Capital preservation over maximum returns")
print(f"• Predictable and consistent performance")
print(f"• Lower volatility in monthly returns")
print(f"• Better sleep at night with stable investments")

# Visualization
plt.figure(figsize=(15, 10))

# Plot 1: Monthly returns comparison
plt.subplot(2, 2, 1)
plt.plot(months, portfolio_a, 'o-', label='Portfolio A', linewidth=2,
        ↪markersize=6)
plt.plot(months, portfolio_b, 's-', label='Portfolio B', linewidth=2,
        ↪markersize=6)
plt.axhline(y=mean_a, color='blue', linestyle='--', alpha=0.7, label=f'Mean A_
        ↪({mean_a:.2f}%)')
plt.axhline(y=mean_b, color='orange', linestyle='--', alpha=0.7, label=f'Mean B_
        ↪({mean_b:.2f}%)')
plt.xlabel('Month')
plt.ylabel('Return (%)')
plt.title('Monthly Returns Comparison')
plt.legend()
plt.grid(True, alpha=0.3)

# Plot 2: Box plot for distribution comparison
plt.subplot(2, 2, 2)
portfolio_data_melted = portfolio_data.melt(id_vars=['Month'],
        ↪var_name='Portfolio', value_name='Return')

```



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portfolio_data_melted['Portfolio'] = portfolio_data_melted['Portfolio'].str.
    ↪replace('Portfolio_', 'Portfolio ')
sns.boxplot(data=portfolio_data_melted, x='Portfolio', y='Return')
plt.title('Return Distribution Comparison')
plt.ylabel('Return (%)')

# Plot 3: Risk-Return scatter
plt.subplot(2, 2, 3)
plt.scatter(std_a, mean_a, s=200, label='Portfolio A', alpha=0.7)
plt.scatter(std_b, mean_b, s=200, label='Portfolio B', alpha=0.7)
plt.xlabel('Risk (Standard Deviation %)')
plt.ylabel('Return (Mean %)')
plt.title('Risk-Return Profile')
plt.legend()
plt.grid(True, alpha=0.3)

# Plot 4: Histogram of returns
plt.subplot(2, 2, 4)
plt.hist(portfolio_a, alpha=0.7, label='Portfolio A', bins=8, edgecolor='black')
plt.hist(portfolio_b, alpha=0.7, label='Portfolio B', bins=8, edgecolor='black')
plt.xlabel('Return (%)')
plt.ylabel('Frequency')
plt.title('Return Distribution Histograms')
plt.legend()

plt.tight_layout()
plt.show()

print(f"\n" + "="*60)
print("FINAL SUMMARY")
print("="*60)
print(f"For a risk-averse investor, {recommended_portfolio} is the better choice,
    ↪because:")
print(f"• Lower volatility ({min(std_a, std_b):.4f}% vs {max(std_a, std_b):.4f}%
    ↪standard deviation)")
print(f"• More consistent returns (variance: {min(variance_a, variance_b):.6f}
    ↪vs {max(variance_a, variance_b):.6f})")
print(f"• Better risk-adjusted performance (lower coefficient of variation)")
print(f"• Minimal sacrifice in expected returns for significantly reduced risk")

```

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INVESTMENT PORTFOLIO RISK ANALYSIS
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MONTHLY RETURNS DATA:

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	Month	Portfolio_A	Portfolio_B
0	1	3.2	4.1
1	2	2.8	3.5

2	3	4.5	5.0
3	4	3.7	4.2
4	5	2.9	4.5
5	6	4.0	3.8
6	7	3.1	4.4
7	8	4.2	3.9
8	9	3.3	4.6
9	10	2.7	4.3
10	11	4.1	4.0
11	12	3.6	4.7

TASK 1: MEAN CALCULATION AND PERFORMANCE ANALYSIS

Portfolio A Mean Return: 3.5083% per month
Portfolio B Mean Return: 4.2500% per month
Difference: 0.7417% (Portfolio B advantage)

ANNUALIZED RETURNS (Compounded):

Portfolio A: 51.25% per year
Portfolio B: 64.78% per year

PERFORMANCE ANALYSIS:

- Portfolio B has higher average monthly returns (4.25% vs 3.51%)
- Portfolio B provides 21.1% higher returns on average
- Over 12 months, Portfolio B outperformed A by 13.53% annually
- Both portfolios show positive returns across all months

TASK 2: VARIANCE CALCULATION AND CONSISTENCY ANALYSIS

POPULATION VARIANCE (N):

Portfolio A Variance: 0.327431
Portfolio B Variance: 0.162500

SAMPLE VARIANCE (N-1):

Portfolio A Variance: 0.357197
Portfolio B Variance: 0.177273

MANUAL VARIANCE CALCULATION (Portfolio A):

Step 1 - Calculate deviations from mean:

Month 1: $3.2 - 3.5083 = -0.3083$
Month 2: $2.8 - 3.5083 = -0.7083$
Month 3: $4.5 - 3.5083 = 0.9917$
Month 4: $3.7 - 3.5083 = 0.1917$
Month 5: $2.9 - 3.5083 = -0.6083$
Month 6: $4.0 - 3.5083 = 0.4917$
Month 7: $3.1 - 3.5083 = -0.4083$

Month 8: $4.2 - 3.5083 = 0.6917$
 Month 9: $3.3 - 3.5083 = -0.2083$
 Month 10: $2.7 - 3.5083 = -0.8083$
 Month 11: $4.1 - 3.5083 = 0.5917$
 Month 12: $3.6 - 3.5083 = 0.0917$

Step 2 - Square the deviations:

Month 1: $(-0.3083)^2 = 0.095069$
 Month 2: $(-0.7083)^2 = 0.501736$
 Month 3: $(0.9917)^2 = 0.983403$
 Month 4: $(0.1917)^2 = 0.036736$
 Month 5: $(-0.6083)^2 = 0.370069$
 Month 6: $(0.4917)^2 = 0.241736$
 Month 7: $(-0.4083)^2 = 0.166736$
 Month 8: $(0.6917)^2 = 0.478403$
 Month 9: $(-0.2083)^2 = 0.043403$
 Month 10: $(-0.8083)^2 = 0.653403$
 Month 11: $(0.5917)^2 = 0.350069$
 Month 12: $(0.0917)^2 = 0.008403$

Step 3 - Sum of squared deviations: 3.929167

Step 4 - Divide by N: $3.929167 / 12 = 0.327431$

CONSISTENCY ANALYSIS:

- Portfolio B is more consistent (lower variance: 0.162500)
- Portfolio A is less consistent (higher variance: 0.327431)
- Variance difference: 0.164931
- Lower variance indicates more predictable returns
- Higher variance suggests more volatile performance

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TASK 3: STANDARD DEVIATION CALCULATION AND RISK COMPARISON

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POPULATION STANDARD DEVIATION:

Portfolio A: 0.5722%
 Portfolio B: 0.4031%

SAMPLE STANDARD DEVIATION:

Portfolio A: 0.5977%
 Portfolio B: 0.4210%

VERIFICATION (variance = std dev):

Portfolio A: $0.327431 = 0.5722\%$
 Portfolio B: $0.162500 = 0.4031\%$

RISK COMPARISON:

- Portfolio B has LOWER RISK (std dev: 0.4031%)
- Portfolio A has HIGHER RISK (std dev: 0.5722%)

- Risk difference: 0.1691%

COEFFICIENT OF VARIATION (Risk per unit of return):

Portfolio A: 16.31%

Portfolio B: 9.49%

- Portfolio B offers better risk-adjusted returns

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TASK 4: RISK ASSESSMENT AND RECOMMENDATION

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COMPREHENSIVE RISK ANALYSIS:

Metric	Portfolio A	Portfolio B	Winner
-----	-----	-----	-----
Mean Return	3.5083	4.2500	B
Variance	0.327431	0.162500	B (lower is better)
Standard Deviation	0.5722	0.4031	B (lower is better)
Coeff. of Variation	16.31	9.49	B (lower is better)
Range	1.8	1.5	B (lower is better)

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RECOMMENDATION FOR RISK-AVERSE INVESTOR:

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RECOMMENDATION: Portfolio B

JUSTIFICATION:

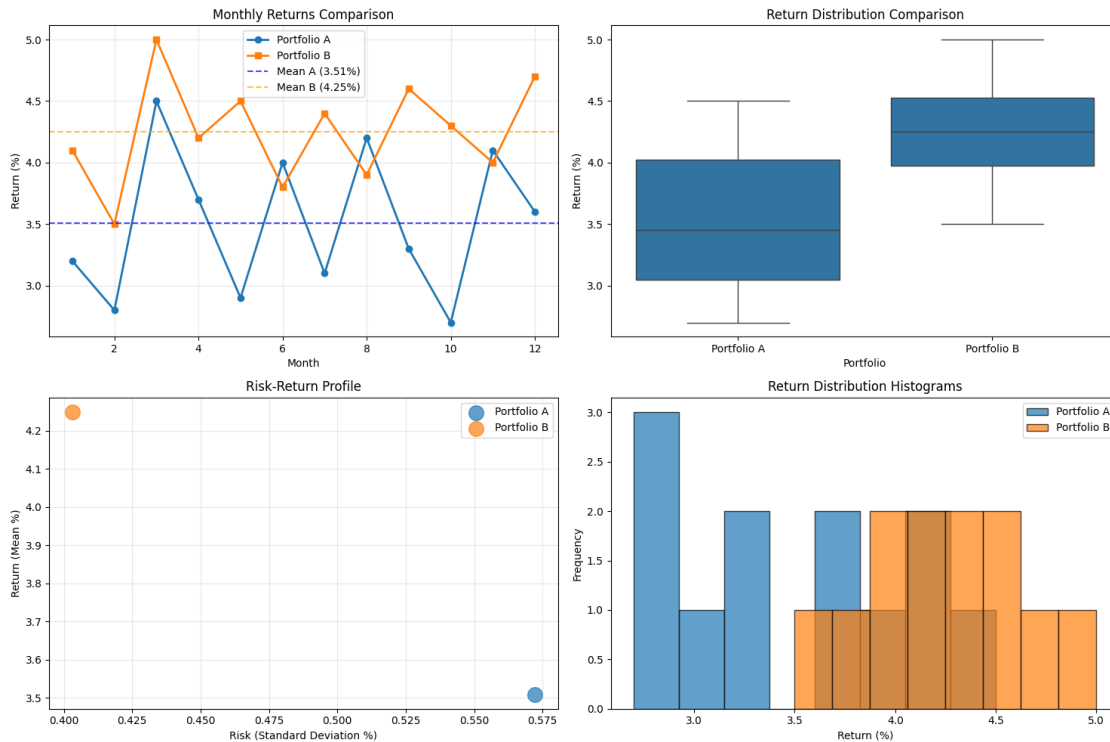
1. ****Lower Risk****: Portfolio B has 0.1691% lower standard deviation
2. ****More Predictable****: Lower variance means more consistent returns
3. ****Risk-Return Trade-off****: Sacrificing only 0.7417% return for significantly lower risk

QUANTITATIVE ANALYSIS:

- Portfolio B standard deviation: 0.4031% (lower risk)
- Portfolio A standard deviation: 0.5722% (higher risk)
- Return sacrifice: -0.7417% per month
- Risk reduction: 0.1691% standard deviation

RISK-AVERSE INVESTOR PRIORITIES:

- Capital preservation over maximum returns
- Predictable and consistent performance
- Lower volatility in monthly returns
- Better sleep at night with stable investments



FINAL SUMMARY

For a risk-averse investor, Portfolio B is the better choice because:

- Lower volatility (0.4031% vs 0.5722% standard deviation)
- More consistent returns (variance: 0.162500 vs 0.327431)
- Better risk-adjusted performance (lower coefficient of variation)
- Minimal sacrifice in expected returns for significantly reduced risk